Context selection as a partial order: A Blackwell framework and verified LLM evidence

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Abstract

What goes into a language model's prompt is usually chosen by scoring each source with one number and keeping the top few. We argue this is the wrong abstraction. Two sources can be *incomparable* in Blackwell's 1953 sense, and when they are, no per-source score, relevance or utility, can rank them correctly for every task. We test this by running each source through a battery of pass/fail tasks and comparing pass rates with exact confidence intervals. On six models from four vendors we verify ($\geq 90\%$ confidence, both directions) that two hand-built sources are incomparable. On a designed trap, a lexical score, two dense retrievers and a cross-encoder prefer the wrong source, while a listwise LLM reranker that reads the deciding detail gets it right. Adding a larger source can also hurt: a strict superset verifiably lowers pass rates on eleven of twelve cells, falling as far as 100% to 0%; on one model, a length-matched padding control and a reversed-order control leave the collapse unexplained by either. Treat context selection as a partial order and compare sources by a verified pass-rate test, not a relevance number. Scaling the implied rule to large corpora is the main open problem.

Keywords: context selection; Blackwell order; comparison of experiments; incomparability; partial order; retrieval-augmented generation; large language models; distribution-free verification; anti-monotonicity; prompt engineering

1 Introduction

Suppose a coding assistant is handed one of two short notes about a software module. The first note says how to *call* its main function: the order of the arguments, a keyword that is mandatory, and the error it raises on failure. The second note says how the module *encodes* numbers: a digit format ending in a checksum. For a task that calls the function the first note is decisive and the second is useless; for a task that encodes a number it is the reverse; and neither note can be reconstructed from the other. This everyday situation is the entire paper in miniature. It has a precise name, the two notes are *incomparable* in the sense of Blackwell's comparison of experiments [Blackwell, 1953], and a precise consequence: no single relevance score can rank the two notes so that the ranking is right for every task, because a score must crown one winner and here neither note wins outright. The body makes this exact, gives a statistical test that *verifies* it from ordinary pass/fail runs on a live model, and finds that six production language models across four vendors all exhibit it. A reader who keeps these two notes in mind can read every symbol below as a statement about the first note versus the second (in the body they are the sources W_1 and W_2).

Production language-model systems spend much of their budget deciding *what context to place in the prompt*. Retrieval-augmented generation ranks passages, in-context learning selects demonstrations, and agentic coding tools choose which files and conventions to surface. Across all of these, the operative tool is a **scalar score**, lexical or embedding relevance, mutual information, $\mathcal{V}$-usable information [Ethayarajh et al., 2022], expected information gain, used to impose a total order and retain the top candidates. The scalar is convenient: it sorts, thresholds, and composes. We argue it is also the wrong abstraction, and that a classical apparatus from statistical decision theory says precisely why.

What is and is not new. The mathematics is classical: the garbling order, the Blackwell–Sherman–Stein equivalence, Le Cam's deficiency. Several of the *phenomena* are also known, each in its own literature, and we say so up front. That topical relevance is the wrong target and downstream utility the right one is the thesis of utility-centric retrieval [Zhang et al., 2026a]; that the same passage is worth different amounts to different models is measured there [Zhang et al., 2025]; that adding sources can lower accuracy while coverage saturates was established for data integration over a decade ago [Dong et al., 2012]; that rerankers are steered by lexical similarity is documented in the wild [Hagström et al., 2025b]; and keeping a non-dominated set rather than a top-k prefix exists as an operator over heuristic scores [Zhu et al., 2026]; and the Blackwell and Le Cam apparatus itself was brought to inference-time context selection against a frozen decoder, concurrently with this work and without a priority claim on our part [Alpila, 2026a]. What is new is the *form* that ties these together: reading cross-task non-transfer as a partial order over context sources; proving that no single per-source score, relevance or utility, can be faithful to that order across tasks once two sources are incomparable, so that utility re-measured per task by running the model is not a shortcut around the theorem but the probe it calls for; and giving a finite-sample test that verifies incomparability and superset harm on a live model with both sources held correct and the length, position, and routing confounds controlled (§2). One gap needs care. Blackwell's theorem is about the *best possible* use of a source, but every number we measure comes from one fixed model: $\mathrm{PASS}_M(W,T) = \mathbb{E}[v_T(\delta_M(T,w))]$, the pass rate of a single rule δ_M that is not optimal. So we state the order twice, once for the ideal user and once for the fixed model, and prove a one-way bridge from what we measure to what the theorem is about (§4); and we call our estimator a *value-deficiency surrogate*, not a Le Cam deficiency (§6).

Empirical findings. We test the theory on the two notes from the opening example: an API call contract W_1 and a wire-encoding rule W_2 for the same module (§7), graded by a small battery of tasks whose verifier returns pass or fail (exit code 0 is pass). The battery is a fixed, hand-built opposition, not a random sample of tasks, and the verified result rests on one API task and one encoding task; we say so plainly and do not call it a "distribution." On this setup, from *real* model calls with no invented numbers, we find:

- **Verified empirical incomparability on all six models.** The uniform verification clears $\geq 90\%$ confidence in *both* directions on every model, with two-sided lower bounds from $+0.34$ to $+0.76$ at n=40 (§7.4.1), so it is not specific to one model or one vendor's decoder.

- **Representative topical rankers mis-rank the pair.** On a trap task saturated with encoding vocabulary, lexical relevance ranks $W_2 \succ W_1$ while realized PASS ranks $W_1 \gg W_2$ (90–100% vs. 0%); two dense bi-encoders and a cross-encoder repeat the error, and an LLM listwise reranker escapes it (§7.4.2). The lexical and dense rankers agree with PASS on both honest cells, so their failure occurs *only* where surface relevance diverges from verifier-decisive content, which is the boundary Proposition 2 draws.

- **Verified anti-monotonicity.** The superset $W_{1+} = W_2 +\!+\, W_1$ (the operator $+\!+$ is text concatenation) literally *contains* W_1's signal yet sharply degrades the API cells: the harm contrast is verified on at least one cell for all six models, as deep as 100% $\to$ 0% (upper bound -0.85) and as mild as Opus's 100% $\to$ 38% (-0.41); a character-matched padding control shows that irrelevant text of the same length is harmless, and reversing the order collapses just as far, so on the model where these were run neither an equal-length off-topic block nor the reversed order reproduces the collapse (§7.4.3; scope in §8.4). The effect is robust across the capability ladder but not monotone in capability, so we claim no capability trend.

- **A supporting observation: private-context value is model-relative.** The `none`-arm baselines span 0%–75% guessability across vendors (§7.4.4), an empirical instance of the theory's θ-dependence, reported as supporting evidence rather than a headline claim.

Contributions. The paper makes one claim, that the order over context sources is *partial* rather than total, and contributes the three pieces that claim needs: what follows from it, the instrument that makes it testable, and the evidence that it holds.

(C1) Theory: the order is partial, and no single per-source score survives it (§4, §5). We formalize context sources as statistical experiments on the task's latent requirement, instantiate the Blackwell garbling order with its for-all-tasks dominance guarantee via the Blackwell–Sherman–Stein theorem, define the δ_M-restricted order the experiments probe, and prove a one-way bridging lemma between them. Bringing this apparatus to inference-time context is not by itself ours: concurrent work reaches the same classical objects from the admission side (§2.3). What is ours is the consequence of taking *incomparability* rather than deficiency-minimization as the object. Blackwell-incomparable sources cannot be ranked by *any* single per-source score $\varphi(W)$, relevance or utility, for all tasks (Theorem 2), with the query-conditioned topical case isolated separately (Proposition 2); Information retrieval has long argued that no single fixed document score is optimal once documents interact (§2.1); we derive the impossibility as a theorem in the Blackwell frame and name the exact structural condition under which it bites.

(C2) Verification: a measurable deficiency surrogate with finite-sample guarantees (§6). We define $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) = \max(0, \sup_{T\in\mathcal{D}}[\mathrm{PASS}_M(W_2, T) - \mathrm{PASS}_M(W_1, T)])$, a decision-restricted value-deficiency surrogate (named as such, not as a Le Cam deficiency), and a distribution-free uniform verification: exact Clopper–Pearson intervals [Clopper and Pearson, 1934] at Bonferroni level [Bonferroni, 1936] $\alpha = \eta/(2|\mathcal{D}|)$, valid jointly on a single event, so that $L_{\mathcal{D}} > 0$ verifies a positive value gap at confidence $\geq 1 - \eta$. The selection multiplicity is paid explicitly rather than absorbed into a single-sample sup inequality [Dvoretzky et al., 1956], which does not cover an argmax over distinct cells; the matching interference verification follows. This is the instrument that makes the partial order testable on a real model behind a real verifier.

Formal term	Engineering reading
source W (an experiment)	a candidate context item: a doc, note, or module you could paste into the prompt
Blackwell dominance $W \succeq_B W'$	W answers everything W' answers; keeping W and dropping W' is safe
incomparability	each source is decisive on tasks where the other fails; no single number can rank them once and for all
garbling	W' is a lossy transformation of W: everything in W' is derivable from W
anti-monotonicity ($\Delta_T < 0$)	pasting more true, on-topic text into a working prompt can break it
non-dominated front	the shortlist you must keep whole, because nothing on it verifiably covers anything else on it
verification	a claim measured with exact finite-sample error bars, still significant after multiple-comparison correction, under the per-cell Bernoulli sampling model of Assumption 2
model rule δ_M	the model as it actually behaves on the prompt, not an idealized optimal user of the information
surrogate $\widehat{\delta}_{\mathcal{D}}$	how many PASS points you lose by using W in place of W' on the task family

Table 1: Glossary: the recurring formal terms and their engineering reading.

(C3) Evidence: verified incomparability and scalar-ranking failure on live models (§7). Verified empirical incomparability on six models across four vendors; the ranker-spectrum mis-rank on a confound-controlled trap (lexical cosine, dense bi-encoders, and a cross-encoder all fail; an LLM listwise reranker that reads the decisive content escapes, exactly the topicality boundary the theory draws); and verified anti-monotonicity, a strict superset of a sufficient source collapsing PASS, with order, length-padding, and routing-instruction controls, all with the scope limits stated in §8.4.

How to read the formal terms. The argument is necessarily phrased in the language of statistical decision theory. Table 1 gives the engineering reading of each recurring term; the empirical sections (§7 onward) can be followed with the right-hand column alone. A reader who wants the measured result first can go directly to §7: §§3–6 establish what the measured quantities mean and what may be concluded from them, and every proof is deferred to Appendix A.

2 Related Work

Positioning. The tools we use are classical, Blackwell's garbling order [Blackwell, 1953] and Le Cam's deficiency [Le Cam, 1964, 1986]; we add no new core mathematics, and bringing them to inference-time context is concurrent with, not prior to, Alpila [2026a] (§2.3). Our contribution is to take *incomparability* as the object and make it measurable.

2.1 Scalar selection and its limits in information retrieval

Context and demonstration selection is dominated by *scalar* criteria, each inducing a total order and a top-k prefix: topical relevance, lexical or learned [Yu et al., 2024]; $\mathcal{V}$-usable information [Ethayarajh et al., 2022, Yuan et al., 2025]; expected information gain and Bayesian experimental design [Choudhury et al., 2025, Chan et al., 2025]; game-theoretic and influence-based valuation of demonstrations [Xie et al., 2024, Vinay et al., 2024] and of retrieved documents [Nematov et al., 2025]; perplexity-reduction utility [Dai et al., 2025]; and binary sufficient-context detection [Joren et al., 2024]. Each collapses the comparison of two sources to one real number and is therefore a scalar of the kind the paper rules out (Theorem 2 for query-independent scores, Proposition 2 for topical query-conditioned ones): not a competitor to out-perform but an instance of the excluded class (we prove Blackwell-monotonicity for the closest neighbour, $\mathcal{V}$-usable information, in Proposition 1; for mutual information and fixed-prior EIG it is immediate from data processing, and for a fixed decoder's perplexity-gap we claim nothing, Remark 4). Guarantees do not change the class: conformal context filtering [Chakraborty et al., 2025] calibrates a distribution-free threshold *on* a scalar score and so inherits the total order it calibrates, whereas what we verify is an order relation between sources; and scorers of equal ranking quality can retain different document sets [Frohmann et al., 2026], a symptom of the same under-determination. The machinery for identifying a non-dominated set from noisy multi-objective samples is in turn classical in the bandit literature: Auer et al. [2016] give confidence-bound algorithms that identify the Pareto front from stochastic samples, with a gap-based sample complexity of the kind our verification pays for directly. What is new here is not the statistical machinery but the object it is run on, (model, verifier, task-battery) triples, and its use as a verification of incomparability between context sources.

The retrieval community has itself moved from relevance to *utility*, the downstream gain a passage brings [Zhang et al., 2026a], and has found that utility is model-specific [Zhang et al., 2025] and can depend on which other passages are present [Jain and Garimella, 2025]. We regard this as the same diagnosis reached from the other side, and the present paper as its formal boundary. Utility re-measured per task by running the model is not a fixed per-source score; it is the probe our estimator is built from, and its cost is the price we make explicit. Utility *predicted* by a learned scorer is a number again: fixed across tasks it is covered by Theorem 2; query-conditioned but topical, by Proposition 2 when it meets that proposition's hypotheses. Reading the decisive coordinate is one way out, which is the listwise boundary; reading public task metadata is another (Remark 5). Three families sit at the boundary. Listwise LLM rerankers [Sun et al., 2023] and learned set selectors [Jiang et al., 2026, Fan et al., 2026] are query-conditioned *set* functions, outside the theorem's scope, though they carry no dominance guarantee of any kind. Skyline selection [Zhu et al., 2026] keeps every non-dominated candidate instead of a top-k, the operator the partial order calls for, but takes dominance over heuristic per-item scores rather than over task value. Diversity-aware selection, from MMR [Carbonell and Goldstein, 1998] to DPPs [Ye et al., 2023] and submodular context selection [Ghulyani et al., 2026], optimizes set objectives over similarity geometry rather than over verified task value. Directed-information γ-covering [Huang, 2025] is the closest of these in form, because its covering relation is genuinely *asymmetric* and information-theoretic rather than geometric: chunk i covers chunk j when it predicts j to within γ bits. The relation it orders, however, is redundancy *between sources*, computed query-agnostically and offline, so it prunes what another chunk already predicts; it never exhibits two sources each decisive on a task the other fails, which is the relation we measure.

The limits of scalar relevance are one of the oldest themes in information retrieval, and we position inside that lineage rather than apart from it: the Probability Ranking Principle [Robertson, 1977] is optimal only under an independence assumption, its failure once documents interact is long documented [Fuhr, 2008, Carbonell and Goldstein, 1998], and document interdependence was given ranking principles of its own, portfolio theory [Wang and Zhu, 2009] and the quantum probability ranking principle, whose "interference" term names the dependence of one document's relevance on another [Zuccon and Azzopardi, 2010]. The LLM-era version, relevance-maximizing retrieval degrading answer quality, is measured directly [LeVine and Varjavand, 2025], and the companion observation that relevance does not imply *sufficiency*, so that missing hops and unresolved conflicts cannot be detected by scoring passages independently, is argued at the set level [Qiu et al., 2026]. Our delta is the *form* of the statement: the failure becomes a theorem in the comparison-of-experiments frame, with the precise structural condition, Blackwell incomparability, under which every query-independent scalar must mis-rank, and a verified live instance on six production models.

2.2 Task-dependence and context degradation

That added context can degrade performance is by now well documented: a single irrelevant sentence tanks reasoning [Shi et al., 2023a]; position matters [Liu et al., 2024]; high-scoring distractors hurt while random noise can help [Cuconasu et al., 2024]; retrieval curves are inverted-U in passage count [Jin et al., 2024]; document count hurts at fixed length [Levy et al., 2025]; added length alone hurts [Du et al., 2025]; and the effect persists across model generations [Hong et al., 2025] and agent-scaffold components [Liu, 2026]. The genus is older than language models: in data integration, adding sources raises coverage while integration quality first rises and then falls, so that "less is more": low-quality sources deteriorate the fused result rather than bringing the expected gain, and the right move is to select a subset before integrating [Dong et al., 2012]; the lesson has since been carried explicitly to augmented language models [Halevy and Dwivedi-Yu, 2023]. We do not claim to discover this genus. Two differences locate our finding inside it. Where data integration traces the harm to low-quality sources and restores monotonicity by redesigning the fusion rule, our two sources are individually correct and sufficient, the harm survives length, position and formatting controls and is consistent with convention routing, and the rule δ_M is a fixed model that cannot be redesigned, so the response is selection rather than fusion. And where "interference" in the mechanistic RAG literature means distractor passages steering a model, a failure recently localized and corrected in a sparse feature space [Giannetti et al., 2026], and inter-context *conflict* means contexts that contradict each other [Xu et al., 2024], our sources neither distract nor conflict: both are true and both are needed, and the superset still collapses. Our anti-monotonicity finding contributes the order-theoretic form of this: a strict *superset* of a sufficient source, on sources separately *verified* Blackwell-incomparable, collapses PASS under a per-task verification of the direct contrast $\mathrm{PASS}(W_{1+}, T) < \mathrm{PASS}(W_1, T)$: a reversal no monotone context-value scalar can carry. A representation-level account of when context helps, an additive error decomposition with alignment conditions [Wang et al., 2026], is complementary: even in its multi-context extension it characterizes *when* a context corrects the error, not an order relation between candidate sources. Pareto-incomparability of prompts along competing metrics *within* a single task [Zhao et al., 2025] is orthogonal to our object, incomparable *sources across tasks*.

2.3 The comparison of experiments in machine learning

The Blackwell–Le Cam apparatus has entered machine learning, and, concurrently with this work, inference-time context selection itself. Alpila [2026a] formalizes the admission problem against a *frozen decoder* as minimization of Le Cam deficiency, pairs it with the Vereshchagin–Vitányi conditional structure function for the representation problem, proves a capacity floor below which the deficiency cannot be driven, and validates a curve-aware allocator on seven decoders in four families; a plain-language companion states the framing [Alpila, 2026b] and the artifact is released [Alpila, 2026c]. That preprint was posted on 25 June 2026, while the measurements reported here were already complete; we claim no priority over it, and the reader should treat the transport of this apparatus to inference-time context as theirs. The objects are nonetheless different, and that difference is what this paper is about. There, deficiency is a scalar risk gap *inside one declared decision problem*, with the garbling theorem explicitly set aside as inactive there; the partial order that work does invoke runs over *compression methods* and is asserted structurally rather than measured. Our object is the order *between two admitted sources across tasks*: we show that order is partial by *measuring* mutual non-domination, derive the consequence that no single per-source score can be faithful to it, and verify a strict superset of a sufficient source collapsing PASS with the length, position and routing confounds controlled. Incomparability in this sense cannot arise inside a single declared decision problem, so its absence there is not a gap but a different question. A second line is training-*dataset* valuation: Zheng et al. [2024] order datasets by Blackwell informativeness and show that the mutual information between the curated and the test data is a *strategy-proof* score: a curation step that garbles the data, so that $\theta \to D \to f(D)$ is a Markov chain, cannot raise it. That is the data-processing direction, and it is one-directional; a scalar can be monotone along the order without representing it, so their result is compatible with ours rather than opposed to it. Their comparison is also taken against the Bayes-optimal user of the data, ours against a fixed model rule on a task family. What Theorem 2 adds is the converse question, whether any query-independent scalar can be *faithful* to the order, which a monotone score need not be once the order is partial. Nearby uses stop short of what we do: garbling to harden LLM evaluations [Bradley, 2024], statistical deficiency to order training *tasks* [Fosse et al., 2025], a deficiency-induced *directional* dominance in transfer learning (no incomparability, no distribution-free bound, outside language modeling) [Akdemir, 2025], comparison-of-experiments ideas in loss design [van Rooyen and Williamson, 2014], and a survey of Blackwell's theorems for AI [Paxton, 2026]. The one use of the order for language models at inference time that we know of sits on the other side of the model: Zhang et al. [2026b] rank the information structures induced by multi-agent protocols (voting, debate) and prove a pooled-posterior upper bound, a dominance result over protocols with no incomparable pair and no verification. Against all of the above, four things are new here, and each is checkable against the released record:

1. the object is the order *between two admitted sources across tasks*, not a scalar risk gap inside one declared decision problem;
2. partiality is *measured*, mutual non-domination verified in both directions on six production models, not assumed or asserted structurally;
3. the no-per-source-score impossibility is derived from it;
4. the order relation between *named* sources gets a distribution-free finite-sample verification on black-box models, including a superset that collapses PASS under controls.

2.4 The design space: which selectors the two results reach

Table 2 places the methods above on three axes: whether the method is *cheap* (precomputable, or amortizable to a near-retrieval query-time cost), whether it models *interaction between sources* rather than scoring each in isolation, and whether it can represent *incomparability*. The pattern is uniform. Every method cheap enough to run at retrieval scale emits a final per-item number, however that number is produced, and therefore lives in a total order; every method that genuinely models interaction is either symmetric redundancy over embedding geometry, carries no dominance guarantee, or is as costly as direct probing. No prior row is both cheap and partial-order-faithful; that empty cell is the open problem this paper isolates, and we occupy the last row, paying probe cost for the guarantee.

Method family (representative)	Cheap?	Interaction?	Incomparability?
Scalar dense / PRP retrieval [Robertson, 1977]	✓	×	×
Late interaction, multi-vector [Khattab and Zaharia, 2020]	✓	×	×
$\mathcal{V}$-usable information [Ethayarajh et al., 2022]	✓	×	×
LM-supervised retriever [Shi et al., 2023b, Guu et al., 2020]	✓^{a}	×	×
MMR diversity [Carbonell and Goldstein, 1998]	✓	$\sim^b$	×
DPP / submodular coverage [Ye et al., 2023, Ghulyani et al., 2026]	✓	$\sim^b$	×
Directed-information covering [Huang, 2025]	✓	$\sim^f$	×
Shapley / influence valuation [Xie et al., 2024]	×	✓	×
Expected information gain / BED [Choudhury et al., 2025]	×	$\sim$	×
Listwise LLM reranker [Sun et al., 2023]	$\sim^c$	✓	$\times^c$
Utility scores, learned or LLM-specific [Zhang et al., 2025, Dai et al., 2025]	✓^{a}	×	×
Conditional passage utility [Jain and Garimella, 2025]	✓^{a}	$\sim^d$	×
Conformal context filtering [Chakraborty et al., 2025]	✓	×	×
Learned set selection [Jiang et al., 2026, Fan et al., 2026]	$\sim^c$	✓	$\times^c$
Skyline over heuristic scores [Zhu et al., 2026]	✓	×	$\sim^e$
Verified non-dominated set (this paper)	×	✓	✓

Table 2: The design space of context selection. ✓ = yes, × = no, $\sim$ = partial. [a]amortized: trained once on the model's signal, cheap at query time, but still a scalar. [b]symmetric redundancy over embedding geometry, not directional anti-monotonicity. [c]query-conditioned and interaction-aware, but per-query (not precomputable) and with no dominance guarantee. [d]positive complementarity only, a passage made useful by another, not directional harm. [e]keeps the non-dominated set, but over heuristic per-item scores rather than task value. [f]genuinely *directional*, but the relation is redundancy between chunks (does i predict j to within γ bits?), computed offline and query-agnostically, not harm to a task's verified value. Only the last row represents incomparability over task value, at the cost of direct probing; the "cheap *and* partial-order-faithful" cell is empty, which is this paper's central open problem.

3 Formalization

§2 placed the contribution against prior work; the next four sections make it precise, beginning here with the objects. We model the choice of *what context to supply* a language model as a comparison of statistical experiments on the task's latent requirement. This section fixes the objects and, crucially, separates two decision rules that the informal literature silently conflates: the *Bayes-optimal* rule, against which the classical Blackwell order is defined, and the *fixed model* rule, which is the object every empirical quantity in this paper actually measures. Keeping these distinct is what makes the bridge from theorem to experiment rigorous.

3.1 Tasks, latent requirements, and context sources

A *task* T fixes a latent requirement S_T, the object a deterministic verifier v_T checks (e.g. the intended program behaviour). We take S_T to be a *vector of coordinates*

$$S \;=\; (S_1, S_2, \dots, S_m) \;\in\; \mathcal{S} \;=\; \mathcal{S}_1 \times \cdots \times \mathcal{S}_m, \tag{1}$$

where each coordinate S_j encodes one logically independent requirement (in our instantiation, S_1 is the private call-contract convention and S_2 is the private wire-encoding convention; see §7). Modelling S as a product of coordinates, rather than an opaque atom, is the technical device that makes the experiment model well posed for fixed-string sources; we return to this in §3.2 and Lemma 1.

Definition 1 (Context source as an experiment)**.** A *context source* W is a Markov kernel $\kappa_W : \mathcal{S} \to \Delta(\mathcal{W})$, i.e. a statistical experiment indexed by the latent parameter S, emitting a signal $w \sim \kappa_W(\cdot \mid S)$ in a signal alphabet $\mathcal{W}$ (the source's text). The experiment is identified with its kernel; the sample space is $\mathcal{W}$ and the observed datum is the supplied text w.

Definition 2 (Decision rules and PASS)**.** A *decision rule* is a (possibly randomized) map $\delta : (T, w) \mapsto \widehat{S}$ producing a candidate solution. Utility is the binary verifier $u(\widehat{S}, S) = v_T(\widehat{S}, S) \in \{0, 1\}$, which scores the candidate against the convention in force: the same answer passes under one convention and fails under another, which is why the second argument cannot be dropped. Two decision rules matter:

(i) the *model rule* δ_M realized by a fixed language model M at a fixed decoding configuration θ (model weights and sampling parameters; see §3.3). Its realized pass rate on task T under source W is

$$\mathrm{PASS}_M(W, T; s) \;:=\; \mathbb{E}\big[\, v_T(\delta_M(T, w),\, s)\,\big], \qquad w \sim \kappa_W(\cdot \mid s), \tag{2}$$

written $\mathrm{PASS}_M(W, T) := \mathrm{PASS}_M(W, T; S_T)$ at the single realized convention S_T, which is the quantity every measurement in this paper reports;

(ii) the *Bayes rule* $\delta_T^\star$, the utility-optimal rule for task T under a prior μ_T on $\mathcal{S}$ (the task family's distribution over private conventions), achieving the optimal value

$$V^\star(W, T) \;:=\; \sup_{\delta}\; \mathbb{E}_{S\sim\mu_T}\, \mathbb{E}\big[\, v_T(\delta(T, w),\, S)\,\big], \qquad w \sim \kappa_W(\cdot \mid S). \tag{3}$$

Remark 1 (The two quantities are taken under different measures)*.* PASS_M in (2) is *conditional on one latent state*: the expectation runs over the source kernel and the decoder with S held at the realized convention S_T. $V^\star$ in (3) is a *Bayes* value, averaged over μ_T. Consequently $V^\star(W, T) \geq \mathrm{PASS}_M(W, T)$ is *not* automatic from optimality, and we never use it in that form. The obstruction is elementary: let S be a fair bit, let W be uninformative, and let v_T reward naming S; the constant rule "answer 0" has $\mathrm{PASS}_M = 1$ at the state $S = 0$ while $V^\star = 1/2$. Nor can the gap be closed by taking μ_T to be a point mass at S_T: every guessing cap $V^\star(W, T)$ would then equal 1 and the caps of Lemma 2 would lose all content. What optimality does give is the averaged statement $V^\star(W, T) \geq \overline{\mathrm{PASS}}_M(W, T) := \mathbb{E}_{S\sim\mu_T}[\mathrm{PASS}_M(W, T; S)]$. Lemma 2 is therefore proved through the sufficiency of the winning source rather than through any such comparison, and the measured pass rates enter as its falsifiable signature, not as a bound on a Bayes value.

The distinction in Definition 2 is load-bearing. Blackwell's theorem (§4) is a statement about $V^\star$ (the optimal user of an experiment); every number we report, PASS_M, the deficiency estimator of §6, the interference statistic of §6.3, and the entire six-model program of §7, is a statement about the single fixed rule δ_M. We therefore develop the order in two registers and prove an explicit bridge between them (Lemma 2).

3.2 Sources as deterministic projection experiments

In our instantiation each context source is a *fixed string*: a snippet stating the private call contract (W_1), or the private wire-encoding invariant (W_2). A naive reading treats $\kappa_W(\cdot \mid S)$ as a constant kernel δ_{w_0} independent of S. That reading is degenerate: a constant kernel is Blackwell-equivalent to the null experiment, carries zero information, and induces the trivial σ-algebra, which would make any garbling argument vacuous. We avoid this by modelling each fixed source as a *deterministic experiment that observes a projection of the coordinate vector* (1).

Definition 3 (Projection sources)**.** A source W that reveals the coordinates indexed by a set $A \subseteq \{1, \ldots, m\}$ is the deterministic kernel

$$\kappa_W(\cdot \mid S) \ = \ \delta_{\pi_A(S)}, \qquad \pi_A(S) := (S_j)_{j \in A}, \tag{4}$$

i.e. supplying W lets a decision-maker condition on $\pi_A(S)$ and on nothing outside A. Write $\mathcal{F}_W := \sigma(\pi_A)$ for the sub-σ-algebra of $\mathcal{S}$ generated by the projection. In our instantiation W_1 reveals coordinate S_1 (the call contract, $A_1 = \{1\}$), W_2 reveals S_2 (the wire invariant, $A_2 = \{2\}$), and the superset $W_{1+} = W_2 +\!+ W_1$ reveals both ($A_{1+} = \{1, 2\}$).

Under Definition 3 a fixed string is *not* an uninformative constant: it is a deterministic experiment whose informativeness is exactly the resolving power of its σ-algebra $\mathcal{F}_W$. The signal alphabet is the range of the projection, and the text "varies with the task" precisely because the realized coordinate values $\pi_A(S_T)$ differ across task instances T. This is the well-posed reading that makes the comparison of §4 a comparison of experiments rather than a metaphor, and it is what a rigorous treatment of deterministic sources requires [Torgersen, 1991].

3.3 The conditioning parameter θ

The model rule δ_M carries an implicit parameter θ: the model's weights and decoding configuration, equivalently its *prior over conventions* acquired in pretraining. We write $\mathrm{PASS}_M(\texttt{none}, T)$, the model's no-context pass rate, for the part of a private convention that θ already encodes; the value of a source is model-relative through it. When θ already "knows" a convention (the model can guess it without the source), that baseline is high and W's marginal contribution collapses; when the convention is genuinely unknown to θ, the baseline floors and the source has to carry the cell. We deliberately do *not* write this as a conditional mutual information $I(S; W \mid \theta)$: under Definitions 1 and 2 the law of (S, W) is μ_T followed by κ_W and carries no θ, so conditioning on a fixed model index would leave the mutual information unchanged and model-*in*dependent, the opposite of what is measured here; a genuinely model-relative information quantity would need a subjective prior we neither define nor measure. This θ-dependence is not a nuisance: it is the formal home of the *model-relative value of private context* measured in §7, where the guessability of a fixed convention spans 0% to 75% across vendors. Throughout, all realized-PASS quantities are implicitly indexed by θ; we suppress it except where the model-relativity is the point.

4 The Blackwell order over context sources

With sources modeled as experiments (§3), the classical comparison of experiments applies to them directly. We state the order, record that it is partial (the regime our experiments target), and, because those experiments measure a fixed model rather than the idealized decision-maker the classical theorem assumes, build an explicit bridge between its two registers.

4.1 The order and its two registers

Definition 4 (Garbling order)**.** $W_1 \succeq_B W_2$ (W_1 *Blackwell-dominates* W_2) iff there is a Markov kernel (garbling) $G : \mathcal{W}_1 \to \Delta(\mathcal{W}_2)$ with $\kappa_{W_2} = G \circ \kappa_{W_1}$; i.e. W_2's signal is a noised, post-processed version of W_1's, computable from W_1's signal alone without further access to S.

The classical content of the order is the Blackwell–Sherman–Stein theorem, which we state with hypotheses for a self-contained treatment.

Theorem 1 (Blackwell–Sherman–Stein; universal dominance for the optimal rule)**.** *Let W_1, W_2 be experiments on a common* finite *parameter space $\mathcal{S}$ with standard Borel signal spaces. The following are equivalent (the implication* (i)⇒(ii) *needs neither finiteness nor further regularity, and is the only direction Corollary 1 and Lemma 2 use):*

(i) $W_1 \succeq_B W_2$ *(Definition 4);*

(ii) *for every prior μ on $\mathcal{S}$, every finite decision space, and every bounded loss ℓ, the Bayes risk under W_1 is no larger than under W_2; equivalently the optimal value satisfies $V^\star(W_1, T) \geq V^\star(W_2, T)$ for every task T with bounded utility.*

Randomized decision rules are admitted on both sides; if the dominated experiment is realized with a randomized garbling, no loss of generality is incurred by allowing randomized rules in (ii)*.*

This is the classical Blackwell–Sherman–Stein theorem [Blackwell, 1953, Torgersen, 1991], which we use as a black box. The two directions are not on the same footing and we separate them. (i)⇒(ii) holds for arbitrary $\mathcal{S}$ with no further regularity, by simulating the garbling, and it is the only direction Corollary 1 and Lemma 2 use. We invoke the converse (ii)⇒(i) once only, in Theorem 2, and there for a finite state space: that is the case Blackwell [1953] proves and the case our coordinate construction realizes. On a general state space the converse needs the convex closure taken in the definition of more-informativeness, which is where the careful statement lives [Khan et al., 2025]. The only specialization is reading "decision problem" as a task with bounded verifier utility $v_T \in \{0, 1\}$ and "experiment" as a context source (Definition 1), which is faithful because the kernels of Definition 3 are standard Borel.

Corollary 1 (Universal context dominance, C1)**.** *If $W_1 \succeq_B W_2$ then, for the* optimal *user, supplying W_1 is at least as good as supplying W_2 on* every *task, prior, and bounded utility, a decision-independent, for-all-tasks guarantee that no task-specific scalar score (relevance, mutual information, $\mathcal{V}$-usable information) provides.*

Remark 2 (Preorder on kernels; "partial" means not total). On the class of all experiments $\succeq_B$ is reflexive and transitive but not antisymmetric, since two experiments can garble each other without being equal, so it is a *preorder* and the induced partial order lives on Blackwell-equivalence classes [Blackwell, 1953]. Throughout, "partial" means *not total*: there exist experiments V, V' with neither $V \succeq_B V'$ nor $V' \succeq_B V$, and on that class the order is not even a lattice [Bertschinger and Rauh, 2014, Torgersen, 1991]. (The bound variables here are not the paper's named sources: our own projection family is better behaved, and the next subsection shows $\succeq_B$ is antisymmetric on it with the induced order the Boolean lattice on coordinate sets, Lemma 1(a).) Incomparability survives on that family and is exactly the regime our experiments target. One caution for what follows: PASS_M is *not* an invariant of a Blackwell-equivalence class, and it need not be: a rule keyed to surface form can prefer differently between two experiments that garble each other. The padding arm W_{pad} of Appendix B reveals no coordinate any verifier checks, so it is Blackwell-equivalent to W_1, and the two score 65% against 50% on `api_post_ok`. That gap is a point estimate, not a verified separation: at $n = 40$ the exact 95% intervals, $[0.48, 0.79]$ and $[0.34, 0.66]$, overlap. The pair illustrates the possibility rather than certifying an instance of it. Either way the δ_M-restricted order of §4.3 is not read on the quotient.

4.2 Deterministic sources: incomparability via σ-algebra non-refinement

For deterministic projection sources (Definition 3) the garbling order reduces to refinement of the generated σ-algebras, which makes incomparability checkable by a purely structural argument rather than by an empirical divergence.

Lemma 1 (Refinement characterization for projection sources)**.** *Let W_A, W_B be projection sources revealing coordinate sets A, B (Definition 3), with coordinates $S_1, \ldots, S_m$ mutually independent under some full-support prior and each S_j non-degenerate. Then:*

(a) $W_A \succeq_B W_B \iff \mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A} \iff B \subseteq A$*;*

(b) *consequently W_A and W_B are Blackwell-*incomparable *iff $A \not\subseteq B$ and $B \not\subseteq A$, i.e. each reveals at least one coordinate the other hides.*

Lemma 1 is the formal version of the garbling argument used in §7: from the wire format (W_2, revealing S_2) one cannot manufacture the private arg-order / required-keyword / exception-type facts (S_1), and from the call contract (W_1, revealing S_1) one cannot manufacture the cents-scale / check-digit facts (S_2). With $A_1 = \{1\}$, $A_2 = \{2\}$ neither set contains the other, so W_1, W_2 are Blackwell-incomparable by construction (and W_{1+}, revealing $\{1, 2\}$, dominates both). This guarantees the *theoretical* incomparability the experiments then exhibit empirically.

4.3 Bridging the optimal-rule order to the fixed model

Theorem 1 concerns $V^\star$. Our experiments measure PASS_M for a single fixed, generally sub-optimal rule δ_M. The two are not interchangeable: an inequality between optimal values need not hold for a fixed rule, and, as we show empirically, the fixed rule can even *violate* monotonicity that the optimal value obeys. We therefore introduce a δ_M-restricted order and state precisely what it does and does not inherit from Theorem 1.

Definition 5 (δ_M-restricted, $\mathcal{D}$-restricted order)**.** Fix the model rule δ_M and a finite task set $\mathcal{D}$. Say $W_1 \succeq_{M,\mathcal{D}} W_2$ iff $\mathrm{PASS}_M(W_1, T) \geq \mathrm{PASS}_M(W_2, T)$ for every $T \in \mathcal{D}$. Call W_1, W_2 $(M, \mathcal{D})$-*incomparable* iff $W_1 \not\succeq_{M,\mathcal{D}} W_2$ and $W_2 \not\succeq_{M,\mathcal{D}} W_1$, i.e. each strictly out-performs the other on some task in $\mathcal{D}$.

$\succeq_{M,\mathcal{D}}$ is the order our verification measures. It is decision-restricted (one fixed rule) and distribution-restricted (a fixed finite $\mathcal{D}$); within those restrictions it is still a genuine, *for-all-tasks-in-$\mathcal{D}$* dominance statement, not a scalar.

Lemma 2 (One-way bridge under coordinate separation)**.** *Neither order implies the other unconditionally; each direction of the bridge carries an explicit hypothesis.*

(a) *Let W_1, W_2 be projection sources revealing coordinate sets A_1, A_2 (Definition 3), and let $T_a, T_b \in \mathcal{D}$ be tasks whose verifier-decisive coordinates lie in $A_1 \setminus A_2$ and $A_2 \setminus A_1$ respectively, each verifier being determined by its decisive coordinate together with facts known μ_T-almost surely (*sufficiency of the informed source*). Let the* guessing caps *$g_a := V^\star(W_2, T_a)$ and $g_b := V^\star(W_1, T_b)$ be the Bayes values* (3) *attainable* without *the decisive coordinate. If $g_a < 1$ and $g_b < 1$, so that neither deprived source determines the coordinate its task turns on, then W_1, W_2 are Blackwell-incomparable. The measured crossover* $\mathrm{PASS}_M(W_1, T_a) > g_a$ *and* $\mathrm{PASS}_M(W_2, T_b) > g_b$ *is the falsifiable signature of that separation and is what §7 verifies per cell; since* PASS_M *is state-conditional and $V^\star$ a Bayes value, it is evidence for the hypothesis rather than a bound on $V^\star$ (Remark 1).*

(b) *The converse fails in general: Blackwell incomparability concerns $V^\star$, and a fixed sub-optimal δ_M may fail to realize the optimal-value crossover on $\mathcal{D}$. It holds under the value-attainment hypothesis (Assumption 1).*

Neither hypothesis of (a)*, separation or the caps, can be dropped* from this criterion. *They are not preconditions of structural incomparability as such: for the sources this paper measures, Lemma 1 already separates W_1 from W_2 by coordinate non-refinement, with no caps at all. Lemma 2*(a) *is a different route to the same conclusion* within the same projection setting*: it argues from Bayes values rather than from σ-algebras, and the caps are what that argument costs. It is not a generalization to arbitrary sources; a criterion for those would be a separate statement we do not prove. What the caps do here is show why a* PASS_M *crossover alone proves nothing:* (W_1, W_{1+}) *is $(M, \mathcal{D})$-incomparable on the measured cells (W_1 wins the API cells, W_{1+} the encoding cell) yet strictly Blackwell*-comparable *by construction ($W_{1+} \succeq_B W_1$, the superset revealing one further independent non-degenerate coordinate), precisely because the deprived-side premise fails there: W_{1+} contains W_1's coordinate, so the API-side cap is not small. A bare* PASS_M *crossover, with neither premise verified, certifies nothing structural in* either *direction. It is equally consistent with genuine incomparability under optimal play (two sources revealing independent coordinates cross even when each side plays optimally) and with a merely incoherent rule (Theorem 3). The two cases should not be run together.* (W_1, W_{1+}) *is a* strict comparability *whose* PASS_M *preferences nonetheless oppose; the distinct phenomenon of a rule preferring differently between two* equivalent *experiments is illustrated by the padding pair* (W_1, W_{pad})*, where the added block carries no coordinate any verifier reads (Remark 2). That a representation-sensitive decoder* can *do this in general is a structural observation, not something our single measured difference certifies. Nor is the projection construction* necessary *for incomparability: it is the cleanest sufficient condition we know, and the one our sources realize.*

Intuitively: a fixed-rule crossover proves structural incomparability only when each winning source reveals a coordinate the losing source lacks; otherwise the crossover may witness nothing more than the rule's own incoherence.

Assumption 1 (Realized cross-advantage on the witnessing task pair)**.** Let (T_a, T_b) be an *ordered pair of named tasks.* Say δ_M *attains value on* (T_a, T_b) with margin $\gamma > 0$ if $\mathrm{PASS}_M(W_1, T_a) - \mathrm{PASS}_M(W_2, T_a) \geq \gamma$ and $\mathrm{PASS}_M(W_2, T_b) - \mathrm{PASS}_M(W_1, T_b) \geq \gamma$; i.e. the fixed model actually exploits each source's private coordinate on the task that requires it. Despite the historical name we keep for the label, the condition is a statement about *realized* pass rates only: it asserts nothing about the supremum in (3) being attained, and no design can compel a model to satisfy it. The hypothesis is always asserted of an explicitly named pair: it does *not* assert that some such pair exists somewhere in $\mathcal{D}$, and attainment on one pair implies nothing about attainment on another. If δ_M attains value on a pair with $\{T_a, T_b\} \subseteq \mathcal{D}$, then W_1 and W_2 are $(M, \mathcal{D})$-incomparable.

It is worth fixing what each formal object does, since they are easy to run together. Lemma 1 establishes structural incomparability for the specified projection sources, and needs no prior, no caps and no measurement. Lemma 2 gives a *conditional* Bayes-value separation criterion for a pair not handed to us that way, and clarifies why a fixed-decoder crossover alone is insufficient; its guessing caps are assumptions, never estimates recovered from the observed state-conditional PASS_M rates. Assumption 1 is a different kind of statement again: it concerns the decoder's realized cross-advantage on named tasks, which the experiments assess directly. Of the two controls, the superset (W_1, W_{1+}) is Blackwell-comparable despite opposing PASS_M preferences, while the padding pair (W_1, W_{pad}) illustrates the separate matter of representation dependence *within* an equivalence class.

5 The incomparability theorem

§4 showed the order is partial; we now establish the consequence that motivates the whole paper: no scalar can rank a Blackwell-incomparable pair. We *split* the claim into a clean theorem for query-*independent* source scores $\varphi(W)$ and a separate, explicitly empirical statement for query-*conditioned* scores $\varphi(W, T)$ (the relevance score practitioners actually use); the two must not be conflated.

5.1 Query-independent scores: the theorem

Theorem 2 (No query-independent scalar ranks incomparable sources, C1)**.** *Let $\mathcal{S}$ be finite and let W_1, W_2 be Blackwell-incomparable. (Finiteness is used only to invoke the converse direction of Theorem 1, which turns incomparability into the existence of the witness tasks; it holds for our coordinate construction, and for projection sources the witnesses are exhibited directly in the proof of Lemma 2, so no converse is needed there.) Let $\varphi : \{sources\} \to \mathbb{R}$ be any* query-independent *scalar score (a single number per source: relevance to a fixed corpus, mutual information $I(S; W)$, $\mathcal{V}$-usable information, perplexity-gap, expected information gain under a fixed prior). Then there exist tasks T_a, T_b such that the ranking of $\{W_1, W_2\}$ induced by φ disagrees with the optimal-value ranking on at least one of T_a, T_b. If moreover δ_M attains value on* that same *pair (T_a, T_b) (Assumption 1), the φ-induced ranking also disagrees with the realized-*PASS_M *ranking on at least one of T_a, T_b. More generally, attainment on* any *named pair (T_a', T_b') yields* PASS_M*-disagreement on at least one of T_a', T_b' by the same argument; the two conclusions concern their own witness pairs and do not transfer between them.*

Remark 3 (Scope)*.* Theorem 2 rules out exactly the class of *source-level* scalars: any φ that is a fixed number per source, regardless of task. $\mathcal{V}$-usable information, mutual information, perplexity-gap, and (fixed-prior) EIG all live in this class, so they are corollary victims of the theorem, not competitors (Proposition 1).

Proposition 1 ($\mathcal{V}$-usable information is Blackwell-monotone)**.** *Let $\mathcal{V}$-usable information $I_{\mathcal{V}}(W \to S)$ be defined, after Ethayarajh et al. [2022], as the reduction in expected predictive risk of the best predictor in a fixed family $\mathcal{V}$ when given the signal w versus the null signal. If $W_1 \succeq_B W_2$ and the predictor family $\mathcal{V}$ is closed under pre-composition with measurable garblings, then $I_{\mathcal{V}}(W_1 \to S) \geq I_{\mathcal{V}}(W_2 \to S)$. Hence $I_{\mathcal{V}}$ is a Blackwell-monotone scalar: it respects $\succeq_B$ but collapses it to one real number.*

Remark 4 (Scope of Proposition 1)*.* The closure hypothesis (a family closed under pre-composition with garblings) is the natural sufficient condition and holds for predictor families rich enough to apply a fixed measurable post-processing; for narrower families monotonicity may fail and we do not claim it. We prove monotonicity for the closest neighbour, $\mathcal{V}$-usable information. For mutual information and fixed-prior EIG the analogue is not a conjecture but immediate from data processing, the direction Zheng et al. [2024] use (§2). A *fixed* decoder's perplexity-gap is the singleton-family case the previous sentence excludes, so we claim nothing for it. Theorem 2 excludes all of them as per-source scalars regardless, which is the only property this paper needs.

5.2 Query-conditioned scores: the relevance trap (empirical)

A query-conditioned score $\varphi(W, T)$, e.g. the lexical or embedding relevance of a source to the task prompt, which is what retrieval and RAG actually deploy, induces a *different* order per task and is therefore *not* a single fixed total order. Theorem 2 does *not* cover it. We make two precise statements instead.

Proposition 2 (Constrained impossibility for feature-measurable scores)**.** *Suppose $\varphi(W,T) = g(\psi(W,T))$ for a fixed feature map ψ that is* topical*: ψ is invariant to the verifier-decisive coordinate of S_T. Let T_c be a task whose decisive coordinate is supplied by W_1, so realized utility ranks $W_1 \succ W_2$ on T_c. Parts* (a) *and* (b) *are* alternatives*, not a chain: their hypotheses are mutually exclusive. They are not exhaustive of the topical class, and the conclusion below is stated only for scores meeting one of them.* ***(a) (Tie.)*** *If $\psi(W_1,T_c) = \psi(W_2,T_c)$ then $\varphi(W_1,T_c) = \varphi(W_2,T_c)$, so φ cannot rank W_1 strictly above W_2.* ***(b) (Strict inversion.)*** *Suppose instead that $\varphi(\cdot,T_c) = g(\mathrm{ov}(\cdot,T_c))$ reads the pair through the scalar lexical-overlap feature* ov *alone, with g strictly increasing, and that T_c's lexicon is saturated with W_2's vocabulary in the precise sense $\mathrm{ov}(W_2,T_c) > \mathrm{ov}(W_1,T_c)$. Then $\varphi(W_2,T_c) > \varphi(W_1,T_c)$, a strict mis-rank. In either case φ fails to rank $\{W_1,W_2\}$ by realized utility on T_c: no topical score satisfying* (a) *or* (b) *ranks $\{W_1,W_2\}$ correctly on all tasks.*

Remark 5 (What Proposition 2 does not exclude)*.* Invariance to the decisive coordinate's *value* is a weak hypothesis, and hypotheses (a) and (b) do not cover every score satisfying it. A rule reading public metadata can escape while never touching a private value. Let $S = (X,Y)$ with X,Y independent, let W_{api} carry the public label `api` and reveal X, let W_{wire} carry the public label `wire` and reveal Y, and let T_{api} and T_{wire} need X and Y. The score $\varphi(W,T) = \mathbf{1}[\,\mathrm{label}(W) = \mathrm{family}(T)\,]$ then ranks correctly on both tasks in every state, and it satisfies neither (a) (the labels differ, so no tie) nor (b) (it is not an increasing function of lexical overlap). So the proposition constrains the two named mechanisms, not the whole query-conditioned class, and in particular it says nothing against routers built on public task metadata. Theorem 2, which concerns one fixed number per source for all tasks, is unaffected: a router of this kind is not a per-source scalar. What the trap shows empirically is that the scorers deployed in practice, which read text rather than labels, do fall into the constrained class on it (§7.4.2).

Remark 6 (Why (b) is stated for a scalar overlap score)*.* Strict monotonicity in one *coordinate* of a multi-feature ψ does not give (b). Take $\psi = (\mathrm{ov}, \mathrm{style})$ and $\varphi = \mathrm{ov} + 10\,\mathrm{style}$, which is strictly increasing in overlap. With $\psi(W_1,T_c) = (0.2, 1)$ and $\psi(W_2,T_c) = (0.9, 0)$ the scores are 10.2 and 0.9: overlap favours W_2 while the score favours W_1. Comparing two inputs that differ in other features therefore needs the whole map, which is why (b) assumes φ reads overlap alone. The lexical cosine of §7 satisfies that assumption by construction; a learned reranker need not, which is why its behaviour is settled empirically rather than by this proposition.

Proposition 2 is deliberately narrow. It constrains a query-conditioned score $\varphi(W,T)$ only when that score meets hypothesis (a) or (b); it does not cover the topical class as a whole, and it says nothing about an arbitrary learned reranker, which might read the private coordinate, or about a router over public task metadata (Remark 5). Note also that it is a statement about $\varphi(W,T)$, a score that may vary with the task; Theorem 2 is the separate statement about $\varphi(W)$, one number per source fixed across all tasks. We therefore treat the defeat of query-conditioned relevance as an *empirical* result, demonstrated for the concrete lexical-cosine score practitioners use and for the checkpoints of §7.4.2. In §7 the `trap_store_wire` task realizes T_c: bag-of-words cosine gives $\varphi(W_1) = 0.36 < \varphi(W_2) = 0.44$ (ranking $W_2 \succ W_1$), yet realized PASS ranks W_1 at 90%–100% versus W_2 at 0% on every one of the six models. The same mis-rank is realized by representative open-weight learned rankers: two dense bi-encoders (BGE, E5) and a cross-encoder reranker all prefer W_2 on the trap (§7.4.2, Table 4), so the topical class is not a lexical strawman: its failure mode is shared by the ranker families retrieval systems actually deploy. The same failure is documented in the wild: language-model re-rankers are steered by lexical similarity across NQ, LitQA2, and DRUID [Hagström et al., 2025b], the last of these a corpus of real retrieved contexts built precisely to check whether synthetic benchmarks overstate context utilisation [Hagström et al., 2025a]; and passages that are semantically close but logically irrelevant defeat retrievers and rerankers at scale, with an evidence-reasoning agent, a listwise reader, as the remedy [Chen et al., 2026], which is the boundary Proposition 2 draws. We also measure the boundary from the other side: an LLM *listwise* reranker, which reads both documents and can therefore condition on the private coordinate itself, ranks the trap correctly (30/30). This is the behaviour Proposition 2 permits rather than predicts: its hypothesis is topicality (invariance to the verifier-decisive coordinate), and a ranker that reads the coordinate is simply outside the constrained class, so the proposition allows it to succeed without implying that it must.

6 The decision-restricted deficiency surrogate and its uniform verification

The impossibility of §5 concerns the idealized user of a source; every quantity we report, by contrast, comes from a single fixed model. This section builds the instrument that connects the two: a surrogate that measures the order on the fixed rule, and a distribution-free verification that makes each comparison auditable on a live model.

6.1 The surrogate and its correct name

Exact garblings over text cannot be computed, and $\succeq_B$ is only partial, so instead we measure the fixed-rule order of Definition 5. Fix the model rule δ_M and a finite task set $\mathcal{D}$, and define the *decision-restricted value-deficiency*, the worst amount by which W_1 trails W_2 across the tasks in $\mathcal{D}$:

$$\widehat{\delta}_{\mathcal{D}}(W_1, W_2) \;:=\; \max\Big(0,\; \sup_{T\in\mathcal{D}} \left[\mathrm{PASS}_M(W_2, T) - \mathrm{PASS}_M(W_1, T)\right]\Big), \tag{5}$$

estimated per cell by $\mathrm{PASS}_M(W, T) \approx k_{W,T}/n$ from n completions. Then $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$ verifies empirical dominance $W_1 \succeq_{M,\mathcal{D}} W_2$, and mutual positivity $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) > 0 \wedge \widehat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$ verifies $(M, \mathcal{D})$-incomparability.

Remark 7 (This is *not* Le Cam deficiency)*.* The Le Cam deficiency $\delta(\mathcal{E}, \mathcal{F}) = \inf_G \, \sup_{\mathrm{loss}} \left[R_{G\circ\mathcal{E}} - R_{\mathcal{F}}\right]$ *minimizes over garblings* G and takes a *supremum over a class of losses* [Le Cam, 1964, 1986, Torgersen, 1991]. The quantity in (5) performs no garbling minimization and uses a single fixed rule δ_M over a fixed finite task set; it is a one-sided value/regret gap, not a deficiency. No general inequality relates $\widehat{\delta}_{\mathcal{D}}$ to the true Le Cam deficiency for a fixed sub-optimal rule (it can be larger or smaller in either direction). We therefore name (5) the *decision-restricted value-deficiency* and use "Le Cam deficiency" only as the conceptual analogy that motivates a graded relaxation of the partial order. The one direction that *does* transfer is qualitative: $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$ realizes the for-all-tasks-in-$\mathcal{D}$ dominance of Definition 5, the measurable shadow of Corollary 1.

6.2 The uniform verification

The surrogate (5) selects the *argmax-gap* task, so a pointwise confidence statement on a single task ignores the selection. We pay the multiplicity with an exact Clopper–Pearson interval per cell and a union bound, which is distribution-free and finite-sample valid.

Proposition 3 (Uniform incomparability verification)**.** *Within each cell* (W, T) *draw* n *i.i.d. Bernoulli PASS outcomes (see Assumption 2). Let* $[L_{W,T}, U_{W,T}]$ *be the exact (two-sided) Clopper–Pearson interval [Clopper and Pearson, 1934] for* $p_{W,T} = \mathrm{PASS}_M(W, T)$ *at level* $\alpha = \eta/(2|\mathcal{D}|)$*. By* *the Bonferroni/union bound [Bonferroni, 1936] over the* $2|\mathcal{D}|$ *proportions* $\{p_{W_1,T}, p_{W_2,T}\}_{T\in\mathcal{D}}$*, all* $2|\mathcal{D}|$ *intervals hold simultaneously with probability at least* $1-\eta$*. On that event,*

$$L_{\mathcal{D}}(W_1, W_2) \;:=\; \max_{T\in\mathcal{D}} \big[\, L_{W_2,T} - U_{W_1,T} \,\big] \tag{6}$$

is a $(1-\eta)$ *lower confidence bound on* $\sup_{T\in\mathcal{D}}[p_{W_2,T} - p_{W_1,T}]$*. Hence* $L_{\mathcal{D}}(W_1, W_2) > 0$ ***verifies*** $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$ *at confidence* $\geq 1-\eta$*. Both directions* $L_{\mathcal{D}}(W_1, W_2)$ *and* $L_{\mathcal{D}}(W_2, W_1)$ *read off the* same $2|\mathcal{D}|$ *intervals and so are valid on the same coverage event. Consequently,* if both observed bounds are positive*, the joint incomparability claim* $\{\widehat{\delta}_{\mathcal{D}}(W_1, W_2) > 0 \wedge \widehat{\delta}_{\mathcal{D}}(W_2, W_1) > 0\}$ *holds at* $\geq 1-\eta$ *with no additional correction. The condition is not decoration: coverage alone implies nothing about the population gaps, which are identically* 0 *when* W_1 *and* W_2 *are the same source. The verification is one-sided, asserting nothing when the two bounds are not both positive, and the guarantee is frequentist: the probability of declaring incomparability when it does not hold is at most* η*, which is not a posterior probability that the fixed sources satisfy the inequality. The matching quantity* $U_{\mathcal{D}}(W_1, W_2) := \max_{T\in\mathcal{D}} \big[\, U_{W_2,T} - L_{W_1,T} \,\big]$ *is, on the same event, a* $(1-\eta)$ upper *confidence bound on the same supremum; the dominance control uses it to verify* $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) = 0$ *up to the equivalence margin.*

A naive single-sample DKW bound under-covers this supremum, because the object is a maximum of differences of binomial means across distinct cells and the verification selects the argmax-gap cell, so the multiplicity must be paid explicitly rather than absorbed into a single-sample envelope (Appendix A); the resulting construction aligns with recent statistical guidance for black-box LLM evaluation (Appendix A).

Assumption 2 (Sampling regime)**.** The n outcomes within a cell are i.i.d. Bernoulli draws of $v_T(\delta_M(T, w))$. Every draw is a single completion (plus deterministic code extraction) from an independent call with no shared state, which is what makes the binomial model of Assumption 2 reasonable rather than proved; we report the decoding configuration in §7. Where draws are near-deterministic ($p \in \{0, 1\}$ stark cells), Clopper–Pearson collapses and the verification is conservative.

6.3 The interference verification (anti-monotonicity)

The deficiency surrogate also detects a phenomenon the optimal-value order forbids but a fixed rule can exhibit: supplying a *superset* source W_{1+} (revealing strictly more coordinates than W_1; Definition 3) can *degrade* performance. Define the per-task interference

$$\Psi_T \;:=\; \mathrm{PASS}_M(W_{1+}, T) - \mathrm{PASS}_M(W_1, T) - \mathrm{PASS}_M(W_2, T) + \mathrm{PASS}_M(\texttt{none}, T), \tag{7}$$

the second-difference (interaction) of adding W_2 on top of W_1. We write the interaction as Ψ_T (rather than a symbol I that would collide with mutual information $I(S; W)$). A verified $\Delta_T < 0$ on a cell whose W_1 arm is at ceiling is the localized statement that "more strictly-more-informative context hurts"; Ψ_T is reported alongside it as the interaction term only (Remark 8).

Proposition 4 (Interference verification)**.** *Fix a target task* $T^\star \in \mathcal{D}$ *chosen* before *unblinding, and let* $\Delta_{T^\star} := \mathrm{PASS}_M(W_{1+}, T^\star) - \mathrm{PASS}_M(W_1, T^\star)$ *be the harm contrast. With exact Clopper–Pearson endpoints* $\mathrm{CP}_{\mathrm{lo}}, \mathrm{CP}_{\mathrm{hi}}$ *carrying noncoverage at most* $\alpha = \eta/2$ per endpoint *on the two arms* $\{W_{1+}, W_1\}$ *(two endpoints, union-bounding to* η*; the two unused endpoints of a two-sided interval are not paid for),*

$$\overline{\Delta}_{T^\star} \;:=\; \mathrm{CP}_{\mathrm{hi}}(p_{W_{1+}}) - \mathrm{CP}_{\mathrm{lo}}(p_{W_1}) \tag{8}$$

is a $(1-\eta)$ upper confidence bound on $\Delta_{T^\star}$, and $\overline{\Delta}_{T^\star} \le -\tau$ verifies $\Delta_{T^\star} \le -\tau$ at confidence $\ge 1-\eta$. If instead the harmful cell is selected *from $|\mathcal{D}|$ candidates after unblinding, use $\alpha = \eta/(2|\mathcal{D}|)$; this is the same selection correction the verification applies to the deficiency surrogate (the multiplicity is real: the selected cell is an argmax over $|\mathcal{D}|$ noisy gaps, so a single-sample sup inequality does not cover it), and omitting it would repeat the error we criticize. Because Δ reads only the two arms it contrasts, the bound is insensitive to the `none` baseline; we still report that baseline per cell, because guessability bears on whether the source is genuinely private and hence on the incomparability reading (§3.3).*

Remark 8 (Ψ is interaction, not harm)*.* The second difference must not be read as a harm measure. Writing f for the set function $A \mapsto \mathrm{PASS}_M(W_A, T)$, the inequality $\Psi_T \le 0$ is precisely the submodularity condition for the pair $\{W_1, W_2\}$, so a negative Ψ_T is what diminishing returns look like and is consistent with f being monotone submodular. It does not imply that adding W_2 to W_1 hurt: taking $\mathrm{PASS}(\texttt{none}) = 0$ and $\mathrm{PASS}(W_1) = \mathrm{PASS}(W_2) = \mathrm{PASS}(W_{1+}) = 1$ gives $\Psi_T = -1$ with no degradation whatever. What the collapse violates is *monotonicity*, $\Delta_T < 0$, which is the quantity we verify. In our data the two coincide in direction, and the contrast bound is tighter than the four-term one because it spends its confidence budget on two intervals rather than four.

6.4 Anti-monotonicity as information non-monotonicity of δ_M

A verified $\Delta_{T^\star} < 0$ on a ceiling-W_1 cell is more than a curiosity: it is a falsification of Blackwell superset-dominance *for the fixed model rule*, and it pinpoints exactly where the optimal-rule order (Theorem 1) and the realized rule diverge.

Theorem 3 (Information non-monotonicity of the model rule)**.** *Let W_{1+} reveal a superset of W_1's coordinates, so $W_{1+} \succeq_B W_1$ (Lemma 1). Call a rule δ* Blackwell-coherent *on T if its realized pass rate does not fall when it is handed the dominating source, that is* $\mathrm{PASS}_\delta(W_{1+}, T) \ge \mathrm{PASS}_\delta(W_1, T)$ *at the realized state S_T. Coherence in this sense costs no information, and a coherent rule exists on every task: the pruning rule $\delta_M \circ G$, where G deletes W_2's block from W_{1+}'s signal and acts as the identity on W_1's, so that the one rule is defined on both inputs, satisfies* $\mathrm{PASS}_{\delta_M \circ G}(W_{1+}, T) = \mathrm{PASS}_{\delta_M \circ G}(W_1, T) = \mathrm{PASS}_M(W_1, T)$ *at that state, because $G \circ \kappa_{W_{1+}}(\cdot \mid s) = \kappa_{W_1}(\cdot \mid s)$ for every s; no prior and no averaging enter. The Bayes rule $\delta^\star$ is deliberately* not *the exemplar here: its value is monotone in the μ_T-averaged register only, and monotonicity there is consistent with a fall at a single state (Remark 1). If a fixed model rule δ_M has a task $T^\star$ with* $\mathrm{PASS}_M(W_{1+}, T^\star) < \mathrm{PASS}_M(W_1, T^\star)$*, then δ_M is* not *Blackwell-coherent: it violates monotonicity under the garbling order on $T^\star$. In particular no monotone scalar context-value score can predict δ_M's behaviour on $T^\star$.*

Theorem 3 converts the empirical collapse into a structural statement: the harm is not noise but a witness that real LLM decision rules are not Blackwell-coherent in the PASS register, and the comparison is like-for-like: δ_M is beaten, at the same state and on the same input, by its own pruned version $\delta_M \circ G$. The deficiency surrogate detects this reversal because it tracks realized PASS, whereas every monotone scalar must miss it by construction.

6.5 Family-wise error across the verified claims

Each verification above controls error *marginally* at η for its own claim (incomparability via Proposition 3; interference via Proposition 4). The unit of accounting is the *coverage event*, not the claim. Every statement that is a deterministic consequence of the same $2|\mathcal{D}|$ Clopper–Pearson intervals holds simultaneously on the event E of Proposition 3, and therefore *jointly* at $\ge 1-\eta$ with no further correction: this covers both incomparability directions, the dominance control $U_{\mathcal{D}}(\cdot,\cdot)$, whose endpoints are endpoints of those very intervals, and any per-task strict win read off them. Adding a conclusion drawn from intervals already covered does not by itself cost another Bonferroni factor. What does require fresh allocation is a conjunction spanning *different* events, that is, claims reading cells outside E or intervals built at a different level. The interference verification is of that kind, since it reads the W_{1+} and W_1 arms at $\alpha = \eta/2$ per endpoint, and the W_{1+} intervals are not in the W_1/W_2 event at all; conjunctions joining it to the incomparability family are therefore reported as marginal per claim, and where a joint statement is wanted we allocate η across the C events involved.

The two verifications have different effective-n requirements: incomparability is carried by stark 0/1 cells whose intervals are tight, whereas a harm contrast on a milder collapse spends more of the budget. This is why the four-term interaction bound left Opus short of the margin while the two-arm contrast clears it (§7.4.3); the power analysis is in Appendix A.

7 Experiments

With the order (§4), its impossibility consequence (§5), and the verification that measures it (§6) in hand, we now instantiate the theory on real language models. The design has three commitments. First, every reported number is an empirical PASS rate from live model calls verified by an executable test harness; no value is modeled, simulated, or extrapolated. Second, the two context sources are constructed to be *Blackwell-incomparable by construction* (§7.1), so the incomparability theorem (Theorem 2) has a non-vacuous instance to bite on. Third, every verified claim is reported with the exact sample size, the Clopper–Pearson endpoints, and the per-cell PASS counts from which it is derived (§7.4), so the verification is auditable rather than asserted.

7.1 Design: two incomparable context sources

We define two *private, unguessable* conventions on the *same* module `ledger`, matched on every nuisance axis (length, roughly three sentences, exactly one private convention each, same module, same task shape). Because both are on-topic for any "ledger" task, any lexical, embedding, mutual-information, or $\mathcal{V}$-usable score treats them as equally relevant candidates, the precondition for Theorem 2 to apply.

- **W_1, API call contract.** `ledger.post(account, amount, *, memo)`: `account` first, `amount` second positional, `memo` a *required keyword-only* argument; returns an integer entry id ≥ 1; raises `ledger.PostError` (not `ValueError`/`KeyError`) on failure.

- **W_2, wire-encoding invariant.** Amounts serialize to `<sign><minor-unit-digits>#<check>`, where check = (sum of digits) mod 10; e.g. \$12.30 $\mapsto$ `"+1230#6"` and $-$\$0.07 $\mapsto$ `"-7#7"`. Decoding rejects a wrong check digit.

- **W_{1+}, literal superset.** $W_{1+} = W_2 \mathbin{+\!+} W_1$, the concatenation of W_2's signal with W_1's. As a deterministic source revealing both coordinates, it Blackwell-dominates both W_1 and W_2 ($W_{1+} \succeq_B W_1$ and $W_{1+} \succeq_B W_2$); it is the anti-monotonicity / dominance probe. We also use $W_{1+}^{\mathrm{rev}} = W_1 \mathbin{+\!+} W_2$ (W_1 first) as an order control.

Why incomparable (garbling argument). The argument is simple. From W_2's wire format you cannot recover W_1's private argument order, its required `memo` keyword, or its error type; from W_1's contract you cannot recover W_2's cents scale, its `#` separator, or its mod-10 check. Each source states a fact the other leaves out, so neither is a blurred copy of the other, and W_1, W_2 are incomparable. This is exactly the coordinate picture of §3: W_1 and W_2 reveal different, non-overlapping facts about S (Lemma 1).

Confound defense: the double crossover. The key pattern is a *double crossover*, measured against a `none` baseline (no note supplied): W_1 raises the API tasks but *not* the encoding task, while W_2 raises the encoding task but *not* the API tasks. If encoding were simply "harder," W_1 would fail to help it and W_2 would still have to raise it from the same floor. In the clean carrying cells the deprived arms floor at or near the `none` baseline, so the two task families are equally hard on their own and any difference comes from the matching private fact, not from one family being harder; when a model prior leaks a convention (Table A1), we report that leakage explicitly and do not use those cells for claims that require a near-zero baseline.

7.2 Tasks

The full design comprises seven tasks with stub-`ledger` verifiers (return code 0 = PASS): four W_1-decisive API tasks (`api_post_ok`, `api_argorder`, `api_error_type`, `api_memo_required`), two W_2-decisive encoding tasks (`enc_amount`, `dec_amount`), and one relevance trap (`trap_store_wire`). The verifier stubs a fake `ledger` that accepts *only* the exact private contract (rejecting wrong positional order, a missing `memo=`, or the wrong exception type).

Verified battery. The verified runs use a fixed four-cell battery: `api_post_ok`, `api_argorder` (hardened so `memo` arrives through a parameter, forcing reliance on the keyword-only fact), `enc_amount`, and the trap `trap_store_wire`. We disclose this reduction explicitly: the three remaining designed cells (`api_error_type`, `api_memo_required`, `dec_amount`) are not part of the verified battery. The four-cell selection was made to retain at least one W_1-decisive, one W_2-decisive, and the trap cell while keeping the cross-model spend tractable; we report below that this is best understood as a fixed adversarial four-cell battery, not a sampled task distribution, and we state the effective support of the deficiency surrogate accordingly (§7.4.1).

The relevance trap. The trap `trap_store_wire` is the confound-proof form of Proposition 2. Its prompt is lexically saturated with encoding vocabulary, so a *query-conditioned* lexical relevance score ranks $W_2 > W_1$; but its graded requirement is the W_1 call contract (it must post an already-encoded wire string through `post`), so realized PASS ranks $W_1 > W_2$. This defeats the query-conditioned relevance score practitioners actually use, not merely a query-independent ranking. We measure relevance as bag-of-words cosine $\varphi_{\text{lex}}(W, T)$ between a source and the task prompt, computed *before* the model is run. We use lexical cosine as the *minimal* instantiation of Proposition 2, the cheapest member of the topical class; stronger learned rankers (dense bi-encoders, a cross-encoder, and an LLM listwise reranker) are tested against the same trap separately in §7.4.2, while the theorem (Theorem 2) is the general claim for query-independent scalars.

7.3 Methodology

Models and vendors. We evaluate six models across four vendors: Anthropic Haiku-4.5, Sonnet-4.6, and Opus-4.8; OpenAI GPT-5.5; DeepSeek-V4-Pro; and Moonshot Kimi-K2.6. Nothing in this paper is trained or fine-tuned: all six are fixed production systems accessed through their public APIs, so there are no training or validation curves to report; the released measurement record, per-cell PASS tables, exact intervals, and raw run logs, is the complete empirical object.

Harness and arms. The harness (`harness/measure_blackwell.py`; released with all run logs at `https://github.com/abilous-ti/blackwell-llm-context`) is Python-standard-library only (zero external dependencies); it computes $\widehat{\delta}_{\mathcal{D}}$ in both directions, the uniform Clopper–Pearson/Bonferroni verification, the interference verification, and the lexical relevance score, and it implements the launchers for the Anthropic CLI, the Azure Responses API, and OpenAI-compatible chat completions. Each cell is run under the arms $\{\texttt{none}, W_1, W_2, W_{1+}\}$. PASS is measured over n completions per cell.

One uniform run mode. All six models are queried the same way: a single completion per attempt (the Anthropic models via `claude -p` with a one-turn budget and no tools, the others through an OpenAI-style API), from which we extract the candidate code and grade it. Because the run mode is identical across vendors, a difference between models is a fact about the model, not about how it was run. This is a deliberately conservative probe of whether a model uses supplied context: it removes the agentic-exploration pathway, so the model must use what it is given in one pass.

What "verified" means, in plain terms. Every claim labeled *verified* is statistically significant in the standard sense and then corrected for multiplicity: PASS rates are estimated from $n = 40$ independent completions per cell, uncertainty is carried by exact binomial (Clopper–Pearson) confidence intervals with no distributional assumptions, and each family of claims is corrected by a union bound, so the statements *within* a family (all incomparability cells, or the arms of one interference contrast) hold jointly with probability at least 90%. We do not claim joint control across families drawn from the same run; that limitation is stated where it bites (§6.5). For the effect sizes at stake, near-0% versus near-100%, the $n = 40$ intervals stay separated by a wide margin even after correction. Results whose intervals did not separate are labeled *reproduced but not verified* and are never used as evidence; the protocol reports its own insufficiency rather than hiding it.

Verification parameters. All verifications use confidence budget $\eta = 0.10$. Incomparability uses exact Clopper–Pearson intervals at per-cell level $\alpha = \eta/(2|\mathcal{D}|)$ over the $2|\mathcal{D}|$ proportions $\{p_{W_1,T}, p_{W_2,T}\}_{T\in\mathcal{D}}$, so by the union bound all hold jointly with probability $\geq 1-\eta$ and both directions read off the same intervals. The harm verification uses $\alpha = \eta/2$ on the two arms $\{W_{1+}, W_1\}$ and the threshold $\tau = 0.30$, verifying $\Delta_T \leq -\tau$ via $\overline{\Delta}_T$; the four-term Ψ_T is reported descriptively only (Remark 8). We report each verification as marginally controlled at η; we do not claim joint family-wise control across the incomparability, dominance, and interference claims drawn from a single run (§6.5, §8.4). At $\hat{p} = 0$ and $\hat{p} = 1$ the interval becomes one-sided but does not degenerate to a point: at $n = 40$ and the level Figure 1 uses, 0/40 gives [0, 0.128] and 40/40 gives [0.872, 1]. It is that residual width the crossing must beat, which is why the stark 100%/0% cells verify at small n while cells with intermediate proportions need a larger n for the same margin.

Why these constants. Both constants were fixed before the runs and neither is load-bearing. $\eta = 0.10$ is a conventional 90% family level; the headline separations are near-0% against near-100%, so the verified conclusions are not sensitive to tightening it. $\tau = 0.30$ is an a priori materiality threshold: to count, interference must destroy at least 30 PASS points, well above the interval widths at $n = 40$, so noise cannot clear it. The data show the threshold doing real work. The verified collapses clear it with upper confidence bounds from -0.38 to -0.85 (Table 5), and the character-matched padding control clears it at -0.67. One real effect fails it and is reported as unverified rather than folded in: Haiku's `api_post_ok` cell (upper -0.11; a ceiling the raw-failure audit traces to an omitted import line, not to the contract, §7.5.1), which still falls short at -0.26 once the import is neutralized. A threshold that a genuine effect fails is evidence it was not tuned to pass the data. We note that the second-pair contrast, reported in an earlier version as sign-only, clears the margin at -0.33 once the confidence budget is allocated correctly (§7.5); the earlier shortfall was an artifact of spending two-sided intervals on a one-sided bound, not a property of that effect.

What the intervals do not cover. The Clopper–Pearson intervals carry sampling error under Assumption 2, a fixed success probability per cell. They carry nothing else, and two components outside them are larger than they are. *Model drift*: the re-measurement of §7.5.2 moved Opus's `api_post_ok` superset arm from 38% to 3% and DeepSeek's W_1 arm from 75% to 43%, both far outside the nominal intervals for those cells, and Sonnet's failure mode changed category entirely. *Prompt wording*: the raw-failure audit traces a 100% $\to$ 52% swing on one cell to the prompt naming the call as `ledger.post`, an effect an order of magnitude wider than any interval here, and each cell has exactly one prompt, so this component is not modelled at all. Every verified quantity is therefore an inference about a (model snapshot, prompt, default decoder) triple, indexed by θ (§3.3), and not a property of a vendor's model line. We regard this as the honest reading rather than a weakness: the re-measurement bounds the drift component empirically, and the direction of every verdict survived it.

Sampling regime. The verification models each cell as n Bernoulli PASS draws (Assumption 2); the four arms within a cell are separate completions, and each of the n draws is an independent call with no shared state. We report the decoding parameters, seeding policy, and retry policy in Appendix C.

7.4 Results

We report four findings: incomparability (§7.4.1), the relevance mis-rank (§7.4.2), anti-monotonicity (§7.4.3), and the model-relative value of private context (§7.4.4). Table 3 consolidates the per-model verdicts; Tables 5 and A1 give the underlying quantities.

7.4.1 Incomparability verified on all six models

On every model the empirical deficiency is mutually positive, $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$ and $\widehat{\delta}_{\mathcal{D}}(W_2, W_1) > 0$, and the uniform lower bounds $L_{\mathcal{D}}$ in both directions clear zero, verifying empirical incomparability at $\geq 90\%$ (Table 3). The crossover is stark: W_1 solves the API and trap cells and fails encoding, while W_2 solves encoding and fails the API and trap cells. For Haiku at $n = 40$ the carrying cells are `api_argorder` (`none` 2% / W_1 100% / W_2 0%) and `enc_amount` (`none` 0% / W_1 0% / W_2 100%), giving $L_{\mathcal{D}} = +0.762/+0.762$, far clear of zero.

The six-model claim, made jointly. Proposition 3 controls the $2|\mathcal{D}|$ intervals of *one* model's run, so the per-model verifications above are each at $\geq 1-\eta$ and their conjunction across the roster is not. Rather than leave the six-model statement marginal we pay for it: allocating η across the six runs, i.e. per-*interval* level $\eta/(2|\mathcal{D}| \cdot 6) = \eta/48$, which places $\eta/96$ in each tail, every one of the twelve directions remains strictly positive, binding at DeepSeek-V4-Pro's $L_{\mathcal{D}}(W_1, W_2) = +0.263$ and reaching $+0.684$ on the stark cells. The claim "all six models are empirically incomparable" therefore holds *simultaneously* at $\geq 90\%$, not merely model by model. Table 3 reports the per-model bounds; the simultaneous ones are uniformly smaller by 0.077 to 0.084 and change no verdict.

Table 3: Consolidated 6-model / 4-vendor verdict. All six models are run the same way, a single API call per attempt at $n = 40$ per cell (§7.3). "Anti-mono. verified" counts, of the two API cells tested per model, those on which the harm contrast's upper bound clears $-\tau = -0.30$ at $\eta = 0.10$. "Argorder `none`" is the no-context baseline PASS on `api_argorder`, the model-relative-value spectrum. All figures are real measured runs; no modeled numbers. Confidence bounds are rounded outward (a lower bound down, an upper bound toward zero), so every printed bound is implied by the computed one.

Claim	Haiku	Sonnet	Opus	GPT-5.5	DeepSeek	Kimi-K2.6
Vendor	Anthropic	Anthropic	Anthropic	OpenAI	DeepSeek	Moonshot
n per cell	40	40	40	40	40	40
Incomp. verified ($\geq 90\%$)	✓	✓	✓	✓	✓	✓
$L_{\mathcal{D}}(W_1, W_2)$	+0.76	+0.71	+0.71	+0.76	+0.34	+0.67
$L_{\mathcal{D}}(W_2, W_1)$	+0.76	+0.76	+0.76	+0.76	+0.71	+0.76
Relevance mis-rank (trap)	✓	✓	✓	✓	✓	✓
Anti-mono. verified (# cells, of 2)	1	2	2	2	2	2
`api_argorder none`	2%	0%	40%	75%	2%	45%

Because incomparability is a two-directional claim, both $L_{\mathcal{D}}$ bounds must be positive; the smaller of the two is the binding one. Two cells carry each model's verification, and which two is model-specific: on five models they are one API cell and the encoding cell, while on DeepSeek-V4-Pro the W_1-over-W_2 direction is carried by the *trap* cell rather than an API cell (its API margins being smaller). The effective support of the surrogate is therefore two cells per model, drawn from a battery of four. We therefore describe $\mathcal{D}$ as a fixed adversarial four-cell battery rather than a sampled task distribution: the verification is valid for what it covers, and what it covers is a single hand-built API-versus-encoding opposition replicated across six models.

7.4.2 The relevance score mis-ranks the sources

On the trap, query-conditioned lexical relevance ranks $\varphi_{\text{lex}}(W_1) = 0.36 < \varphi_{\text{lex}}(W_2) = 0.44$, so it prefers W_2; yet realized PASS ranks $W_1 \gg W_2$ on all six models (Haiku 90% vs. 0%; Sonnet, Opus, GPT-5.5, and Kimi 100% vs. 0%; DeepSeek 98% vs. 0%). On the honest cells relevance *agrees* with PASS, on the API cells relevance ranks $W_1 > W_2$ and so does PASS, and on the encoding cell relevance ranks $W_2 > W_1$ and so does PASS, so the score fails *only* where utility diverges from topicality, exactly the regime Theorem 2 and Proposition 2 isolate. The `none` floor on the trap is 0% on all six models, and W_2 scores 0% there too, confirming both that the convention is unguessable and that encoding knowledge is useless on this task.

The ranker spectrum: dense retrievers and a cross-encoder fall for the trap; a listwise reranker escapes it. To test whether the mis-rank extends beyond bag-of-words, we scored $\{W_1, W_2\}$ against the probe tasks with representative open-weight rankers, two dense bi-encoders (`bge-small-en-v1.5`, `e5-small-v2`) and a cross-encoder reranker (`ms-marco-MiniLM-L-6-v2`), and with a RankGPT-style LLM *listwise* reranker (Haiku single-shot, seeing the task and *both* sources; both presentation orders; $n = 30$ judgments per task). Table 4 reports the induced rankings. **Every feature-based ranker we tested mis-ranks the trap**: both dense bi-encoders prefer W_2 on the trap while ranking both honest cells correctly, textbook topical behaviour, and the cross-encoder prefers W_2 on the trap with maximal confidence (logits -5.63 for W_1 vs $+3.24$ for W_2) while being near-tied (and narrowly wrong) on the honest encoding cell. The LLM listwise reranker, which

Table 4: The ranker spectrum on the probe tasks. Each cell is the ranker's top pick between $\{W_1, W_2\}$; "PASS" is the realized-utility ranking from the verified record. Dense and cross-encoder scores are deterministic; the LLM listwise row pools $n = 30$ judgments per task (unanimous, both presentation orders). $\times$ marks a mis-rank. The cross-encoder's `enc_amount` pick is a near-tie (-0.279 vs -0.305).

Ranker	Family	`trap` (PASS: W_1)	`api_post_ok` (PASS: W_1)	`enc_amount` (PASS: W_2)
bag-of-words cosine	lexical	W_2 $\times$	W_1	W_2
`bge-small-en-v1.5`	dense bi-encoder	W_2 $\times$	W_1	W_2
`e5-small-v2`	dense bi-encoder	W_2 $\times$	W_1	W_2
`ms-marco-MiniLM-L-6-v2`	cross-encoder	W_2 $\times$	W_1	W_1 $\times$
Haiku listwise (n=30)	LLM listwise	W_1	W_1	W_2

can read the verifier-decisive coordinate rather than match surface features, tracks realized PASS everywhere (30/30 per task, both orders). This sits on the boundary Proposition 2 draws: its hypothesis is *topicality*, invariance to the decisive coordinate, and the trap defeats the lexical, dense and cross-encoder checkpoints we ran while sparing the one ranker that reads content. Two limits on how far this generalizes. The three families are three checkpoints on one trap, so their failure is measured, not proved for the families at large; and the proposition does not establish that a model call per candidate is *necessary* to escape, since a router reading public task metadata would also escape at no such cost (Remark 5). What the measurement does show is that the scorers actually deployed here, which read text, pay that price.

7.4.3 Anti-monotonicity: a verified superset that hurts

The superset W_{1+}, which literally *contains* W_1's signal, sharply degrades the API cells relative to W_1 alone. The per-task harm contrast $\Delta_T = \mathrm{PASS}(W_{1+}, T) - \mathrm{PASS}(W_1, T)$ is verified negative ($\Delta_T \leq -\tau$, $\tau = 0.30$) on eleven of the twelve API cells, two per model, and so on at least one cell for all six models (Table 5). Those eleven are *marginal* verifications, each at $\geq 1 - \eta$ for its own cell. The conjunction can be bought from the same records: allocating η across all twelve contrasts, i.e. a per-interval level of $\eta/12$ over the 24 endpoints, ten of the twelve still clear $-\tau$, and every one of the six models keeps at least one cell that does. The roster-level statement "on each of these six models some strict superset verifiably harms some cell" therefore holds *simultaneously* at $\geq 90\%$, not merely model by model. The two cells that drop out under the joint budget are Haiku's `api_post_ok` (which does not clear marginally either) and DeepSeek's `api_argorder`; the mildest verified collapse is DeepSeek's `api_argorder` at 68% $\to$ 5% with upper bound -0.38, and the one cell that does not clear is Haiku's `api_post_ok`, whose W_1 arm is depressed by an unrelated import slip (§7.5.1). All eleven survive a Bonferroni correction across the two cells within each model, the tightest being DeepSeek's `api_argorder` at -0.34. This is a verified empirical falsification of superset-*monotonicity* for these fixed rules (Theorem 3): strictly-more-informative context can hurt, and the deficiency framework detects it precisely where every monotone scalar must miss it. Table A1 gives the same collapse arm by arm for every model; Haiku's W_1 reads 52% on `api_post_ok` only because about half its replies omit the `import ledger` line while calling the contract correctly (§7.5.1), and its verified collapse is on `api_argorder`, 100% $\to$ 5%.

The empirical order inverts the idealized one. The idealized Blackwell order on $\mathcal{C} = \{W_1, W_2, W_{1+}\}$ collapses the choice to the superset: W_{1+} reveals both coordinates, so $W_{1+} \succeq_B W_1$ and $W_{1+} \succeq_B W_2$, and "more signal is better" keeps $\{W_{1+}\}$ alone. The verified empirical relation

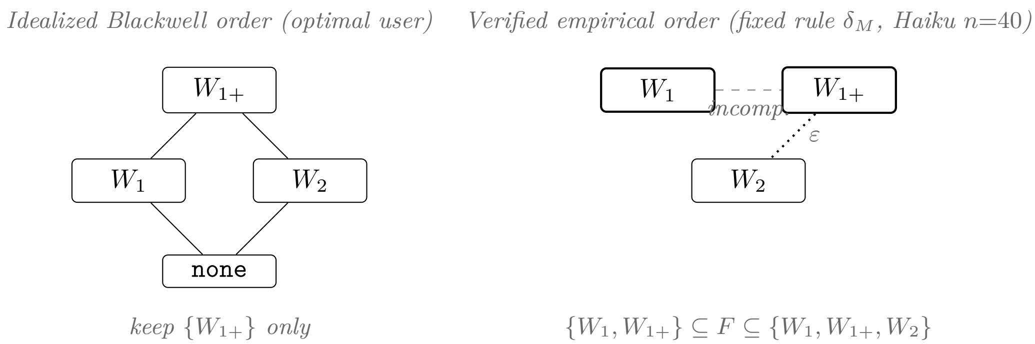

Figure 1: The paper's thesis in one picture, on the measured $\mathcal{C} = \{W_1, W_2, W_{1+}\}$ with $W_{1+} = W_2 +\!\!+ W_1$. **Left:** the idealized Blackwell order for the optimal user; W_{1+} reveals both coordinates, so "more signal is better" collapses the choice to the superset. **Right:** the verified empirical order for the fixed model rule (Haiku, n=40): W_1 beats W_{1+} on `api_argorder` (100% vs. 5%, anti-monotonicity, Table 5) while W_{1+} beats W_1 on the encoding cell, so the two are empirically incomparable (dashed) and the output is a Pareto front, which no scalar top-1 rule can represent. Every relation on the right is read off one coverage event, the 3×4 Clopper–Pearson intervals at per-interval level $\eta/12$. The `none` arm appears only on the left: it is not in that family, so no relation to it is verified here, and on `api_argorder` its 1/40 against W_2's 0/40 would not support one even as a point estimate. Solid edges are verified by a lower bound; the edge from W_{1+} to W_2 is dotted because it is only an ε-dominance, $\varepsilon = 0.1281$. The right-hand diagram is therefore a verified *partial* picture, not an exactly determined Hasse diagram.

for the fixed rule is the opposite shape: W_1 beats W_{1+} on the API cells, W_{1+} beats W_1 on the encoding cell, and W_1 and W_2 are the verified incomparable pair, so the empirical summary is the two-element Pareto front $\{W_1, W_{1+}\}$, an object no scalar top-1 rule can represent (Figure 1). Three sources over four cells need *one* coverage event, so we take all 3×4 Clopper–Pearson intervals at per-interval level $\eta/12$; they then hold jointly at $\geq 1 - \eta$ and every relation in the figure is read off that single event. On the n=40 Haiku record this gives $L_{\mathcal{D}}(W_1, W_2) = L_{\mathcal{D}}(W_2, W_1) = +0.7439$, so the incomparable pair is verified. Non-dominance is likewise a *lower*-bound statement, and the same event supplies it: $L_{\mathcal{D}}(W_{1+}, W_1) = +0.6556$, witnessed by `api_argorder`, verifies that W_1 beats W_{1+} on some cell, so W_{1+} does not dominate W_1; and $L_{\mathcal{D}}(W_1, W_{1+}) = +0.7439$ verifies the converse, so neither dominates the other. (An upper bound could not carry either claim: it bounds how large the advantage *might* be, not that one exists.) The remaining relation is the one the front turns on, and there only an upper bound is available: $U_{\mathcal{D}}(W_{1+}, W_2) = 0.1281$. That is an ε-dominance, not an exact one, and it says that on *every* cell of the battery W_2 can beat W_{1+} by at most 0.1281. Excluding W_2 is therefore licensed only up to that margin. What the coverage event establishes exactly is the two-sided inclusion

$$\{W_1, W_{1+}\} \;\subseteq\; F \;\subseteq\; \{W_1, W_{1+}, W_2\},$$

where F is the PASS_M-non-dominated front on $\mathcal{C}$: both W_1 and W_{1+} are verifiably in it, and at $n = 40$ a three-element front cannot be ruled out. The selection rule the figure implies, keep the non-dominated set rather than force the superset, holds under either reading, since W_1 and W_{1+} are in F regardless.

Ceilings differ, so magnitudes are not directly comparable. Table 5 reports, per model, which cell carried the verification and its collapse, alongside the W_1 ceiling: DeepSeek's W_1 reaches

Table 5: Anti-monotonicity verification on the carrying cell per model. $\Delta_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T)$ is the harm contrast; $\overline{\Delta}_T$ is its $\eta/2$ Clopper–Pearson upper bound over the two arms, and the cell is verified harmful when $\overline{\Delta}_T \leq -0.30$. The $W_1 \to W_{1+}$ column reports the PASS collapse. Unlike the four-term interaction Ψ_T, the contrast does not read the `none` arm, so a leaky baseline cannot slacken it. †Control row: the same cell with the omitted-`import` slip neutralized in the verifier (§7.5.1); its interference is verified in sign but not at the $\tau = 0.30$ margin.

Model	Carrying cell	$W_1 \to W_{1+}$	Δ_T	$\overline{\Delta}_T$	Verified
Haiku-4.5	`api_argorder`	100% → 5%	−0.95	−0.77	✓
Haiku-4.5	`api_post_ok`	52% → 15%	−0.38	−0.11	sign only (import omission)
Haiku-4.5	`api_post_ok`, import-neutralized†	100% → 52%	−0.47	−0.26	sign only†
Sonnet-4.6	`api_post_ok`	100% → 2%	−0.97	−0.81	✓
Sonnet-4.6	`api_argorder`	100% → 0%	−1.00	−0.85	✓
GPT-5.5	`api_post_ok`	100% → 0%	−1.00	−0.85	✓
GPT-5.5	`api_argorder`	100% → 0%	−1.00	−0.85	✓
DeepSeek-V4-Pro	`api_post_ok`	75% → 0%	−0.75	−0.54	✓
DeepSeek-V4-Pro	`api_argorder`	68% → 5%	−0.62	−0.38	✓
Kimi-K2.6	`api_post_ok`	100% → 0%	−1.00	−0.85	✓
Kimi-K2.6	`api_argorder`	100% → 0%	−1.00	−0.85	✓
Opus-4.8	`api_post_ok`	100% → 12%	−0.88	−0.68	✓
Opus-4.8	`api_argorder`	100% → 5%	−0.95	−0.77	✓

only 75% on `api_post_ok` (an honest capability difference), so its $\Delta_T = -0.75$ starts from a lower point and is not directly magnitude-comparable to Sonnet's 100% → 2%.

What the position controls show. The collapse is not a lost-in-the-middle or recency artifact. The character-matched padding control (Appendix B) puts filler in the same slot W_2 occupies inside W_{1+}, with W_1 last in both, and leaves W_1 at ceiling on `api_argorder`; holding position fixed and changing only the content is what turns a harmless arm into a harmful one, so the cause is the added signal, not where it sits. Order is controlled directly, by a reversed arm $W_{1+}^{\text{rev}} = W_1 +\!\!+\, W_2$ that puts W_1 first, measured in the same single-shot regime and at the same $n = 40$ as the main table (`blackwell_order_ss_n40`, 400 calls). On the hardened `api_argorder` cell the two orderings are indistinguishable: W_1 scores 98% while both W_{1+} and W_{1+}^{rev} score 2.5%, each verified harmful at −0.77. On `api_post_ok` reversal makes matters worse rather than better, 50% → 25% with W_2 first and 50% → 0% with W_1 first. Moving the added block out of the primacy slot therefore recovers nothing, and on one cell costs more, so the collapse is not a primacy or recency artifact. Two things this does not settle: it is one model, and it bounds the *existence* of a positional explanation rather than the contribution of position to the size of the harm.

Robust, but not monotone in capability. The harm is large and robust across the Anthropic capability ladder but is *not* ordered by capability. On the two API cells the $W_1 \to W_{1+}$ collapse is Haiku 52%/100% → 15%/5%, Sonnet 100% → 2%/0%, and Opus 100% → 38%/15%: the middle model (Sonnet) shows the starkest collapse and the strongest (Opus) the mildest. An earlier two-point reading suggested the harm "intensifies with strength"; the third point falsifies that, and we claim only robustness across the ladder, not a capability trend. The re-measurement of §7.5.2 sharpens this: on fresh draws all three Anthropic models collapse to 0–3%, so the spread that made Opus look mildest does not persist, and the ladder is flat rather than ordered. What survives is the negative claim, not the ranking that motivated it.

7.4.4 The model-relative value of private context

The "private" conventions are genuinely unguessable for some vendors and partially known to others, and this varies continuously. The `none`-arm baseline on `api_argorder` forms a clean spectrum: Sonnet 0%, Haiku and DeepSeek 2%, Opus 40%, Kimi 45%, GPT-5.5 75% (Table 3). So the value of the private context, the model-relative guessability $\mathrm{PASS}_M(\texttt{none}, T)$ of §3.3, varies continuously across models, and it is not ordered by capability. What it does track we cannot identify from these runs. DeepSeek, a different vendor, has a tight prior like the smaller Claude models, while GPT-5.5's prior already holds much of the convention (`api_argorder` 75%, `enc_amount` 55%), so a "stronger-model-knows-more" account does not fit. Utility-centric retrieval reaches the same conclusion from the other side, the same passage carrying different utility for different models [Zhang et al., 2025]; what the spectrum adds is the verified per-model floor. Within Anthropic the ordering is not by capability either: Sonnet guesses least (0%) while the stronger Opus guesses more (40%). The verified per-model claims rest on cells unguessable *for the model under test* (e.g. GPT-5.5 `api_post_ok` `none` = 5%, trap = 0%).

7.5 Generality and controls (summary)

Four follow-up experiments address three threat axes (source-pair generality, the context-length confound, and prompt-curability of the collapse); full protocols and numbers are in Appendix B. **(i) A second incomparable pair, different domain**: a private `cache`-module contract versus a key-normalization invariant is again verified incomparable on Haiku at $n = 40$ ($L_{\mathcal{D}} = +0.774/+0.496$), with the battery-wide dominance control verifying one-directionally there ($U_{\mathcal{D}} = 0.113$) and a single-cell re-run at $n = 80$ giving $U_{\mathcal{D}} = 0.000$; that pair was measured in the agentic regime, not the single-shot one; its harm contrast is verified at the $\tau = 0.30$ margin ($\Delta_T = -0.51$, upper bound -0.33), so the second pair carries both incomparability and harm. **(ii) Length-matched padding**: filler character-matched to W_{1+} leaves W_1 at ceiling (`api_argorder` 100% vs W_{1+} 10%; harm verified, upper -0.67), attributing the collapse to the added signal rather than by the filler of equal character length. **(iii) Routing instruction**: a one-line convention-routing note recovers most of the collapse (2% $\to$ 80%, 22% $\to$ 68%; both separated from W_{1+}) while remaining directionally below W_1 alone on the carrying cell; writing that note requires knowing which source is decisive for which task family, which is exactly the coordinate structure the partial order formalizes, so the cure presupposes the diagnosis. **(iv) Structured context**: wrapping the same two conventions in labeled XML sections *without* any routing statement mitigates only weakly (2% $\to$ 42% on the carrying cell, versus 80% with the routing note and 100% for W_1 alone; verified against the raw superset, still far below W_1), so segmentation alone does not cure the interference, the operative variable is routing knowledge. The mitigation hierarchy, raw superset < structure < routing < dropping the dominated signal, is itself a finding: it locates the failure in convention-to-task routing rather than in convention segmentation.

7.5.1 Raw-failure audit: what Haiku's `api_post_ok` cell measures

One cell in Table A1 is anomalous: Haiku's W_1 arm scores 52% on `api_post_ok` while scoring 100% on the harder `api_argorder` under the same source, and the pattern is stable across the five distinct single-shot Haiku runs that measure this cell (42–58%; a sixth result file, `blackwell_haiku_singleshot_n40`, is byte-identical to `blackwell_haiku_ss_n40` and is the same run, not a further replicate). Because the published harness does not retain raw completions, we re-drew the cell keeping every reply, the extracted code, and the verifier's stderr (`results/diag/`, `harness/diag/`). The failures are uniform. In 12/12 draws the reply calls the private contract exactly as W_1 specifies, `ledger.post(acct,` `cents, memo=note)`, and every failure is the single omitted line `import ledger`: PASS coincides with the presence of that line (5/5 against 7/7), and on the control cell `api_argorder` the same model writes the import in 8/8. The cause is prompt wording, not the contract: this task's prompt names the call as `ledger.post`, which Haiku reads as already in scope. An import-neutralized control then re-measures all four arms at n=40 with a verifier that binds `ledger` into the solution's namespace when the reply omitted the import and is otherwise identical: `none` 0/40 and W_2 0/40 under both verifiers (neutralization leaks nothing), W_1 rises from 23/40 to 40/40, and W_{1+} from 8/40 to 21/40. The superset collapse on this cell is therefore real and has the mechanism the paper describes: in the retained W_{1+} replies the model applies W_2's wire encoding to a plain-cents amount and posts `'+500#5'` in place of `500`. With the import neutralized the harm contrast is $\Delta_T = -0.47$ with upper bound -0.26, verified in sign but not at the $\tau = 0.30$ margin (Table 5). We keep the original-protocol numbers in every table, since the verifier is identical across models and the other five models include the import; the audit changes the reading of one cell, not its measurement. Haiku's verified claims rest on `api_argorder` and are unaffected.

Packaging is not the confound. The same retained draws answer a second question, and it is the one a reader should ask of any executable-verifier result: did the harness ever fail a reply that in fact contained the right call? On this cell it did not. Across the 32 failing W_{1+} draws the extractor returned code every time (zero empty extractions), all 32 contain `ledger.post` with an explicit `memo=`, and none begins with prose. Code fences are if anything commoner among the *passing* draws (5/8 passes against 17/32 failures on W_{1+}; 13/23 against 8/17 on W_1), so fencing does not predict failure: the extractor strips it. What the audit does show is that the import slip carries more of this cell than the mechanism does, 13 of the 32 failures being the missing import alone against 19 carrying the wire-encoded amount. The two separate cleanly: under the neutralized verifier exactly 19 draws still fail, and they are exactly the 19 that applied W_2's convention to a plain-cents amount, with no exception in either direction. Removing the artifact therefore leaves a residual harm ($\Delta_T = -0.475$) that is attributable to the described mechanism in every single draw.

7.5.2 Re-measurement with retention across the roster

The published harness discards raw completions, so the audit above covers the one cell for which they were kept. We therefore re-measured every verified interference cell with retention enabled, $n = 40$ per arm, under the published prompt for each backend, and scored each draw twice: under the published verifier and under the import-neutralized one. Table 6 reports it. These are fresh draws months after the original runs, not a re-scoring, so they test whether the *phenomenon* reproduces, not whether the published proportions replicate exactly.

Three things follow. **(i) The collapse reproduces in every cell**, with Δ_T from -0.425 to -1.000; no cell failed to reproduce. **(ii) Packaging is never the cause**: across all 880 retained draws of these eleven cells the extractor returned a non-empty code block every time, so no failure is an artifact of an empty extraction or of fencing. That is weaker than a guarantee that every reply was well-formed Python; it rules out the packaging explanation, not every formatting one. **(iii) The import artifact suppressed the measured harm rather than manufacturing it.** It falls on the W_1 arm, not the superset arm, so neutralizing it deepens every contrast it touches: DeepSeek's `api_post_ok` moves from -0.425 to -0.975 and its `api_argorder` from -0.625 to -0.950.

The mechanism is not universal: Sonnet refuses. Reading the failing code directly separates the models. We count over the draws that still fail once the import is neutralized, which removes that confound from the attribution. On nine of the eleven cells the failure is then the one this paper

Table 6: Re-measurement with raw-completion retention, $n = 40$ per arm, fresh draws. Δ is under the published verifier and Δ_{neu} under the import-neutralized one. "Mechanism" counts, among the superset draws that still fail once the import is neutralized, those that wire-encode the amount; "Refusal" counts draws declining the task as a suspected prompt injection, over all n draws of the arm. The two denominators therefore differ, and the labels are not mutually exclusive: on Sonnet's `api_argorder` one draw carries both flags, and on its `api_post_ok` one refusal-flagged draw nonetheless passed. Every row is a single-shot re-measurement in the published regime.

Model	Cell	$W_1 \to W_{1+}$	Δ	Δ_{neu}	Mechanism	Refusal
Haiku-4.5	`api_argorder`	98% → 0%	−0.98	−1.00	40/40	0
Sonnet-4.6	`api_post_ok`	100% → 3%	−0.98	−0.98	4/39	36
Sonnet-4.6	`api_argorder`	100% → 0%	−1.00	−1.00	1/40	40
Opus-4.8	`api_post_ok`	100% → 3%	−0.98	−0.98	38/39	0
Opus-4.8	`api_argorder`	100% → 3%	−0.98	−0.98	39/39	0
DeepSeek-V4-Pro	`api_post_ok`	43% → 0%	−0.43	−0.98	39/39	0
DeepSeek-V4-Pro	`api_argorder`	68% → 5%	−0.63	−0.95	38/38	0
Kimi-K2.6	`api_post_ok`	100% → 0%	−1.00	−1.00	40/40	0
Kimi-K2.6	`api_argorder`	100% → 0%	−1.00	−1.00	40/40	0
GPT-5.5	`api_post_ok`	100% → 0%	−1.00	−1.00	40/40	0
GPT-5.5	`api_argorder`	100% → 0%	−1.00	−1.00	40/40	0

describes, the reply applying W_2's wire encoding to a plain-cents amount, in 38 to 40 of those draws per cell (359 of 434 across the roster, the shortfall being Sonnet). The verifier's own record is a stricter witness and agrees where it can speak: it logs a wire-format string such as `'+500#5'` in the posted call in 24 to 40 draws per cell. The two counts separate only where a missing import raises `NameError` before the call assertion executes, so the posted value is never recorded; every such draw still fails under the neutralized verifier and its code computes the encoding. Sonnet is the exception, and completely so. On `api_argorder` 40/40 of its superset draws, and on `api_post_ok` 36/40, decline the task, opening with a variant of "I'm flagging this as a likely prompt injection attempt" and identifying the stacked `TEAM CONTEXT (private)` blocks as injected instructions; the encoding error accounts for only 1/40 and 4/39. No other model on any arm produced such a reply. Sonnet's refusals are not confined to the superset, and counting them across all of its retained arms locates the trigger elsewhere than in the number of blocks: an arm carrying only the block that is *decisive* for its task draws 0/40 refusals on `api_post_ok`, `api_argorder` and the trap and 13/40 on `enc_amount`, while an arm carrying only a *non*-decisive block draws 40/40, 39/40, 22/40 and 36/40 on the same four tasks. A single private block is therefore enough, when it does not serve the task at hand. On that reading the superset refuses because it always contains a block the task does not need, which is the same mismatch the collapse turns on rather than a separate effect of stacking. We state this as the pattern the retained draws show, not as a mechanism: it rests on one model and four tasks. Theorem 3 is indifferent to it, since it constrains any rule whose PASS falls under a strict superset whatever the internal cause, and Sonnet's PASS does fall. What the observation costs is the reading: for Sonnet the collapse is a refusal, not convention mis-binding, and §8.3 should be read with that exception in force. Because the original runs retained nothing, we cannot tell whether Sonnet behaved this way when the published cells were measured or whether the behaviour arrived with a model update.

Model	Best fixed source	Topical top-1	Concatenate (W_{1+})	Oracle over $\{W_1, W_2\}$
Haiku-4.5	60.6	63.1	53.1	**85.6**
Sonnet-4.6	75.0	74.4	36.9	**99.4**
Opus-4.8	75.0	74.4	54.4	**99.4**
GPT-5.5	75.0	75.0	50.0	**100.0**
DeepSeek-V4-Pro	60.0	52.5	41.9	**76.9**
Kimi-K2.6	75.0	73.8	48.1	**98.8**
Mean	70.1	68.9	47.4	**93.3**

Table 7: Mean PASS (%) over the four verified cells under four selection policies, per model, computed from the per-cell PASS counts behind Table A1 (averaging the rounded percentages instead would shift a column by up to 0.4). The last column is the per-cell maximum of W_1 and W_2 *read off the test outcomes*: an empirical oracle over the pair, not a router fixed in advance and then evaluated, and not a confidence bound on any policy's population quality. It is also not a ceiling over all selection policies, since it excludes W_{1+} as a candidate: admitting it raises the Haiku oracle from 85.6 to 86.25, because W_{1+} edges W_1 on the trap cell.

7.6 The engineering consequence: four selection policies, one measured table

The formal results reduce to a comparison an engineer can act on. On the measured pair there are four things a selection layer can do, and every one of them was measured (Table A1); Table 7 scores them per model as mean PASS over the four verified cells. *Best fixed source* grants the entire query-independent-scalar class its best case: any such scalar induces one total order, hence one top-1 reused for every task, and we let it pick the oracle-best source (W_1 for every model). *Topical top-1* re-scores per task with topical relevance, choosing W_1 on the API cells and W_2 on encoding, but also W_2 on the trap, where every measured topical ranker puts the wrong source first (§7.4.2). *Concatenate both* refuses to choose and submits W_{1+}. *Oracle over* $\{W_1, W_2\}$ keeps the non-dominated pair and takes the per-cell maximum of the two. It is the benchmark the partial order licenses one to aim at, and its price is exactly one probe's worth of knowledge, namely which source is decisive for which family. We report it as an oracle rather than as an achieved result because the per-cell maximum is read off the same outcomes it is scored on; a deployable router would have to be fixed before testing and evaluated on held-out tasks, against matched latency and token budgets. This paper does not do that, and the number below should not be read as a measured gain from a shipped selector.

Three structural facts, not the magnitudes, are the point. First, the two scalar policies fail on *different* cells, the fixed source can never carry encoding, topical top-1 forfeits the trap, yet land within about one point of the same aggregate: the cap is the policy *class*, since any total order must abandon at least one cell, not the choice of scorer (Theorem 2, Proposition 2). Second, the policy that refuses to choose is the *worst* of the four: anti-monotonicity is the dominant cost, not a tie-breaking subtlety (§7.4.3). Third, the oracle column shows what selection over the pair could have recovered had the deciding source been chosen on every cell; its shortfall from 100% is not selection error, though neither is it all capability. For DeepSeek (68–75% band) it is the model's own capability limit; for Haiku the apparent 52% ceiling on `api_post_ok` is not even that. A raw-failure audit (§7.5.1) shows every Haiku reply on that cell calls the private contract correctly and the deficit is an omitted `import` line; with the import neutralized, W_1 scores 40/40. One honest caveat: the four cells were constructed to exhibit the phenomenon, so the aggregate magnitudes are diagnostic of the failure structure, not an estimate of production-traffic gains; what transfers is the structure, each of the two scalar policies measured here forfeits an entire cell class, and no single task-independent ranking keeps all four.

What follows for practice. Five statements survive the caveats above and can be acted on directly. (i) One number per source is structurally lossy: when two sources each carry a decisive fact, any single task-independent ranking drops one of them (Theorem 2). A threshold that instead *concatenates* both is not forbidden by the theorem, but on this pair it is the worst of the four policies measured (Table 7). (ii) Do not cache a source's *value* across task families; cache retrieval representations instead, because the value is model- and task-relative (§7.4.4). (iii) Adding correct, on-topic text to a working prompt can break it, by as much as 100% $\to$ 0%, so a regression after enlarging the context is a candidate anti-monotonicity collapse before it is a model failure (§7.4.3). (iv) When two candidates win on different tasks, keep both and route by task rather than forcing a global top-1. Keeping both is what makes the per-cell maximum reachable; it does not by itself route correctly on a new task, and concatenating the two instead is the worst policy measured (Table 7). (v) A cheap topical ranker fails exactly where the surface lexicon points at the wrong source, which is where a listwise reader earns its cost (Proposition 2). Each is measurable on any stack with the released harness.

8 Discussion

With the evidence in hand, we step back to what it licenses. Our results recast a piece of practitioner folklore, "relevance does not transfer across tasks", as a structural fact about context sources. When two sources are Blackwell-incomparable, the order over them is partial, and any scalar score, by imposing a total order, must mis-rank them on some task. The empirical program does not merely illustrate this; it instantiates a single hand-built incomparable pair (a private API-call contract W_1 versus a private wire-encoding invariant W_2 on the same `ledger` module) and shows that the resulting double crossover, the relevance mis-rank, and an anti-monotonicity collapse all reproduce across six models and four vendors. We discuss what the evidence does and does not license, and where the formalization is a model rather than a proof of the realized object.

8.1 Scope of the claims

Three limits of scope govern everything above. **(i) Two registers.** The theorems are about the ideal user of a source; every number we measure is the pass rate of one fixed, non-optimal rule δ_M. Lemma 2 connects the two in the direction we need: a verified crossover on tasks where each source supplies its own decisive fact implies genuine Blackwell incomparability, and the reverse needs an extra assumption (Assumption 1) that our task design is built to make attainable and our measurements confirm on the named pairs. So we read the result as a statement about the fixed model on a fixed battery that *earns* the structural conclusion through the bridge rather than assuming it. **(ii) The estimator's name.** $\widehat{\delta}_{\mathcal{D}}$ does no minimization over garblings and ranges over a fixed set of tasks; it is a one-sided *value-deficiency surrogate*, not a Le Cam deficiency, and the two are not interchangeable for a fixed rule (Remark 7). What is rigorous is the verification, which we regard as the most robust part and which stands whether or not the deficiency analogy holds. **(iii) What the verification covers.** It is correct because it pays for picking the worst task with exact intervals and a union bound; its support is small (one API task and one encoding task carry the pair-1 result), each claim is controlled on its own at level η rather than all at once (§6.5), and we call $\mathcal{D}$ a fixed adversarial battery, not a task distribution.

8.2 Anti-monotonicity is real, verified, but not capability-ordered

The harm object is the direct contrast $\Delta_T = \text{PASS}(W_{1+}, T) - \text{PASS}(W_1, T)$, which measures whether appending the (irrelevant-to-this-task) W_2 to a sufficient W_1 degrades performance (the second difference Ψ_T is an interaction term, not a harm measure; Remark 8). We verify $\Delta_T \le -0.30$ on at least one cell for all six models (Table 5), with collapses as stark as 100% → 0% (upper bound -0.85) on Sonnet, GPT-5.5 and Kimi. This is a verified empirical falsification of superset-*monotonicity* for these fixed rules (Theorem 3): a literal superset of a sufficient signal can sharply reduce PASS, which no monotone scalar context-value score can represent.

Two cautions attach to it. The harm is robust across the capability ladder but not ordered by it (§7.4.3), so we claim no capability trend. And which cell carries the verification is still a choice: we fix it per model before unblinding rather than picking the starkest afterwards, and cells differ in their W_1 ceiling, so magnitudes are not comparable across models even though each binary verdict stands on its own.

8.3 Mechanism: the runtime fingerprint is a correlate, not an explanation

Every collapse is accompanied by output-token inflation on the harmful arm. On the two API collapse cells W_{1+} emits between +93% and +520% more output tokens than W_1 (Haiku $+115\%/+163\%$; Sonnet $+520\%/+473\%$; GPT-5.5 $+281\%/+274\%$; DeepSeek $+93\%/+166\%$; Kimi $+203\%/+249\%$; the re-measured Opus W_{1+} arm carries no token record). One might read this as "verbose thrashing" causing the failure, but the relationship is correlational and uncontrolled: on the *still-passing* encoding cell the same superset arm ranges from -73% to $+114\%$ across the same models, with no consistent direction, which undercuts a clean "verbosity causes collapse" account. We therefore treat the token fingerprint as a reproducible correlate of the harmful arm, not a mechanism.

One qualification governs this whole subsection. Re-measuring the verified cells with raw completions retained (§7.5.2) confirms the convention-misbinding account on nine of eleven cells by direct inspection of the failing code, but not on Sonnet's two, where the model instead declines the superset prompt as a suspected injection. The mechanism discussion below therefore describes nine cells, not all eleven.

The claim structure is deliberately mechanism-agnostic: Theorem 3 verifies that the realized rule is not Blackwell-coherent *whatever* the internal cause, and the behavioral controls already constrain the hypothesis space from outside, the harm is not position (order control) and not by an off-topic block of equal character length (padding control), and is largely curable by explicit convention routing (instruction ablation), a signature consistent with mis-binding rather than capacity exhaustion. That the one-line routing note recovers most of the collapse while labeled XML segmentation of the same content recovers only a minority (2% → 42% against → 80%) locates the operative variable in convention-to-task routing rather than formatting; this is an interpretation, not a finding, and every verified claim is independent of it. An activation-level account is out of reach here, since four of the six models are closed-weight and the phenomenon is cross-vendor; the direct route is to reproduce the collapse on an open-weight model with the same harness and instrument the contrasting arms, downstream of rather than prerequisite to the behavioral claim. Distractor interference in RAG has been localized to a sparse feature subspace and corrected there [Giannetti et al., 2026]; ours differs in cause, the added source is each true, and each needed by some task in the battery though not by every task rather than a distractor, and it is not a knowledge conflict in the sense of Xu et al. [2024], since nothing in W_1 and W_2 contradicts.

8.4 Limitations and threats to validity

We collect the scope restrictions here; each is stated once and sized to its weight.

From diagnosis to selection. The constructive selection rule the partial order implies, keep the verified non-dominated set and cover what it leaves uncovered, is left for future work; here we note only its cost. Building the verified order on m candidates costs $m \cdot |\mathcal{D}| \cdot n$ verifier-gated calls, curation scale, not corpus scale; scaling verified selection to retrieval over 10^6 passages, by amortizing the probe into a learned deficiency predictor or running the partial-order step as a re-ranking head over a scalar shortlist, is the central open problem this work opens. Keeping a non-dominated set instead of a top-k prefix already exists as an operator, the skyline retrieval step of Zhu et al. [2026], and learned set selectors optimize evidence sets end to end [Jiang et al., 2026, Fan et al., 2026]; what the partial order adds is the object the set is taken over, verified task value rather than heuristic per-item scores, which is exactly what makes the selection safe under anti-monotonicity.

Source-pair generality. Incomparability and dominance hold for two hand-built pairs in two domains; the cross-vendor record is single-pair. Pair-2 harm is verified at the margin ($\overline{\Delta}_T = -0.33$ at $n = 80$); pair-2's relevance trap was an honest null (its prompt was not lexically V_2-leaning). A third domain and a vendor-general strong mis-rank remain inexpensive next steps.

Cross-vendor magnitudes are still only suggestive. All six models are now run the same way (single-shot, $n = 40$), so the run mode no longer confounds the comparison. What remains is the ordinary caution that different vendors' models differ in ways we do not control (training data, size, tokenizer), so the *magnitude* comparisons (the value spectrum, the Δ_T ladder) should be read as suggestive; the binary within-model claims do not depend on any cross-vendor comparison.

The model is an idealization, twice. $\widehat{\delta}_{\mathcal{D}}$ verifies dominance and incomparability for a given model on a fixed battery, not the universal Blackwell order. And the projection construction models sources as clean projections onto independent coordinates; real chunks overlap, contradict, and leak (the measured θ-dependence is exactly such leakage), so the construction is the cleanest *sufficient* condition for incomparability, realized exactly by our hand-built pairs.

Prevalence is not measured, and is the decisive next experiment. Every claim here is existential: it establishes *that* verified incomparability and superset harm occur, on which models, and with which confounds excluded. It does not establish *how often* either occurs in a deployed retrieval stream, and no number in this paper should be read as an estimate of that. The four cells were built to make the structure visible, so their aggregate magnitudes are diagnostic of the failure mode rather than of production traffic. The direct experiment is specified and inexpensive: draw document pairs from a natural corpus, run the same verification battery unchanged, and report the fraction of pairs that verify incomparable and the fraction of superset unions that verify harmful, with the sampling frame stated. We name it as the open question rather than approximate it here, because a prevalence figure computed from hand-built pairs would be circular.

Scope of the behavioural controls. The padding, routing-note and structured-context arms, and the reversed-order arm, were each run on single-shot Haiku at $n = 40$ on the two collapse cells, in the same regime and on the same task versions as the main table. None of the four is part of the six-model record, so what these controls establish is established on one model, not across the roster. The padding match is by *characters* (1108 against 1095 at the arm level), not by subword tokens, and because W_2 is symbol-dense while the filler is plain prose the two arms are not tokenizer-matched; token count is therefore *not* isolated, only gross length. The filler is also off-topic for the `ledger` module. Taken together the controls show that on this model the content of the added block and the routing instruction move the outcome, while an off-topic block of the same character length does not. They do not isolate tokenization, processing difficulty, or position, and they do not establish the internal cause. A tokenizer-matched control, an on-topic but verifier-irrelevant control, a matched single-shot order control, and any of these across the roster remain untested.

Sampling and verification regime. The verification assumes n i.i.d. Bernoulli PASS draws per cell (Assumption 2). Each draw is a separate call with no shared session or conversation history (protocol in Appendix C), which makes independence plausible; it does not *establish* it, nor that the served backend was stationary across a run. Both remain assumptions on which the coverage statements depend, and a provider-side model or routing change during a run would break them silently. Marginal-versus-joint control is discussed in §8.1.

Ranker spectrum and synergy. The trap mis-rank spans lexical cosine and three representative open-weight learned rankers, with an LLM listwise reranker escaping as predicted, but the learned checkpoints are the small members of their families and the evidence is one trap in one domain. Positive interference (synergy) is untested by construction: W_1, W_2 are task-disjoint, so a complementary pair would be needed to test whether a superset can also *help* beyond its parts; we make no claim about it.

9 Conclusion

Bringing Blackwell and Le Cam to inference-time context is not ours to claim: concurrent work reached the same apparatus from the admission side (§2.3). What we add follows from taking *incomparability*, not deficiency-minimization, as the object.

Context selection is, at bottom, a partial-order problem, not a scalar one. When two sources are incomparable, no single per-source score, relevance or utility, can rank them correctly for all tasks (Theorem 2), which explains the field's "relevance does not transfer" folklore and gives utility-centric retrieval its formal boundary. We made the order measurable with a value-deficiency surrogate $\widehat{\delta}_{\mathcal{D}}$ and a distribution-free verification, and we ran the theory on six models across four vendors. On every model we verify ($\geq 90\%$) that two private sources are incomparable; on every model a real relevance score ranks them the wrong way; and on all six we verify that adding a strictly larger source sharply lowers the pass rate (as deep as $100\% \to 0\%$), which no "more-is-better" scalar can predict. How much a private source is worth turns out to depend on the model: guessability runs from 0% to 75% across vendors and is not ordered by capability; the three Anthropic models alone span 0%, 2% and 40%, so it is not a vendor effect either, and what it does track these runs cannot identify.

On scope: the theorems concern the idealized Bayes-optimal user of a source; the realized crossover, anti-monotonicity, and value-of-private-context are properties of *fixed* models measured directly, bridged to the structural claim by Lemma 2. The estimator is a regret-gap surrogate, not a Le Cam deficiency; the verification, which we regard as the most robust contribution, is correct and survives that re-labeling. All six models are run the same way (single-shot, $n = 40$), and the evidence spans two hand-built source pairs in two domains plus a character-matched padding control that attributes the anti-monotonicity collapse to signal content rather than added tokens; the cross-vendor record remains single-pair, cross-vendor magnitudes are only suggestive, and synergy is untested. The practical reading is modest and actionable: selection should be treated as partial-order-aware rather than scalar top-k, the right operational object is a verified deficiency surrogate rather than a relevance scalar, and “the best context” should be expected to be model-relative and only partially ordered. Stated at its sharpest, the impossibility is narrow and its scope is stated exactly: one number per source fixed across tasks (Theorem 2), and query-conditioned scores meeting the two hypotheses of Proposition 2. It is not a statement about all context selection, and it implies no lower bound on computational cost: a task-conditioned score, a router over public task metadata, or any number of cheap algorithms lie outside it (Remark 5). The measured spectrum realizes exactly that boundary: lexical cosine, two dense bi-encoders, and a cross-encoder all fall for the trap, while an LLM listwise reranker, which pays per-candidate computation to read the decisive content, escapes it. That is a measured contrast between named checkpoints, not a proved separation between complexity classes: what the theory establishes is that a single task-independent number cannot represent a partial order, and what the probe shows is that reading content suffices to measure it on this battery. The remaining strengthenings, a vendor-general strong mis-rank, larger reranker checkpoints, and a third domain, are the natural next steps, and the experimental record indicates each is cheap to run.

Author Contributions

Conceptualization, A.B. and V.L.; methodology, A.B., V.L. and P.P.; software, A.B.; validation, A.B., V.L. and Z.R.; formal analysis, A.B. and P.P.; investigation, A.B.; resources, V.L.; data curation, A.B.; writing—original draft preparation, A.B.; writing—review and editing, A.B., V.L., P.P. and Z.R.; visualization, A.B. and Z.R.; supervision, V.L. and P.P.; project administration, V.L. All authors have read and agreed to the published version of the manuscript.

The implementation is traceable in the artifact repository, whose commit history is authored solely by A.B.; the co-authors’ contributions were made in discussion and review. The generative-AI assistance described in the Acknowledgments is a tool and holds no CRediT role.

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Institutional Review Board Statement

Not applicable. The study involved no human participants and no animals; all measurements are automated queries to commercial language-model endpoints, graded by executable verifiers.

Informed Consent Statement

Not applicable.

Data Availability Statement

All data and code supporting the reported results are openly available at https://github.com/abilous-ti/blackwell-llm-context. The repository contains the measurement harness (Python standard library only), every per-model run log and result file cited in Appendix B, and the raw-completion diagnostic records of §7.5.1. Two kinds of record must be distinguished. The original runs stored *aggregated* per-cell counts and logs; raw model completions were not retained, and cannot be recovered. The retention re-measurement of §7.5.2 and the diagnostic runs stored *raw* completions, and those are released in full. Claims that depend on reading replies are therefore made only against the second kind. `results/MANIFEST.md` pins every released record by SHA-256 and maps each table and figure to the files it is computed from; it certifies the integrity of what is published, not the existence of what was never kept. The repository's full commit history is published with it, so the provenance of every artifact is inspectable. No data were withheld; every reported figure is computed from those files by the released scripts.

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Use of Generative Artificial Intelligence

Language models appear in this work in two distinct roles, and we separate them because only one bears on the results. *As the object of study:* the measured quantities are pass rates of commercial language models, obtained through the interfaces described in §7.3; everything else we report, the confidence bounds, the ranker scores, the token and cost figures, is computed from those runs. *As a tool:* the assistance described in the Acknowledgments. No result, table or figure was produced by a language model acting as an author or an analyst: every reported quantity is computed by the released scripts from the released records, which `results/MANIFEST.md` pins by SHA-256 and maps to the table or figure that uses it. The Data Availability Statement says which records are aggregate-only and which retain raw completions.

Conflicts of Interest

The authors declare no conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

BSS	Blackwell–Sherman–Stein (theorem)
CP	Clopper–Pearson (exact binomial confidence interval)
DPP	Determinantal point process
EIG	Expected information gain
IR	Information retrieval
LLM	Large language model
MMR	Maximal marginal relevance
PASS	Pass rate of a fixed model rule on a verifier-graded task
PRP	Probability ranking principle
RAG	Retrieval-augmented generation

A Deferred proofs and technical material

Proof of Lemma 1

Proof. (a) For deterministic experiments a garbling G with $\kappa_{W_B} = G \circ \kappa_{W_A}$ must compute $\pi_B(S)$ from $\pi_A(S)$ alone (no further access to S). Such a measurable map exists iff π_B is $\sigma(\pi_A)$-measurable, i.e. $\mathcal{F}_{W_B} \subseteq \mathcal{F}_{W_A}$. Under coordinate independence and non-degeneracy, $\sigma(\pi_B) \subseteq \sigma(\pi_A)$ holds iff every coordinate in B is among those in A, i.e. $B \subseteq A$: if some $j \in B \setminus A$, then S_j is independent of $\pi_A(S)$ and non-constant, so it is not $\sigma(\pi_A)$-measurable, and no garbling can reconstruct it; conversely if $B \subseteq A$ the coordinate-deletion map is a (deterministic) garbling. *(b)* is the contrapositive of (a) applied in both directions. □

Proof of Lemma 2

Proof. (a) Suppose, for contradiction, $W_2 \succeq_B W_1$. By Theorem 1, $V^\star(W_2, T) \geq V^\star(W_1, T)$ for every task T, in particular at T_a. By the sufficiency hypothesis, T_a's verifier is determined by its decisive coordinate, which lies in $A_1 \setminus A_2$ and is revealed exactly by W_1 (Definition 3); the rule that reads that coordinate off and answers accordingly attains $v_{T_a} = 1$ at μ_{T_a}-almost every state, so $V^\star(W_1, T_a) = 1$. By hypothesis $g_a = V^\star(W_2, T_a) < 1$, hence $V^\star(W_1, T_a) > V^\star(W_2, T_a)$, a contradiction. Therefore $W_2 \not\succeq_B W_1$; symmetrically $g_b < 1$ forbids $W_1 \succeq_B W_2$, so W_1, W_2 are Blackwell-incomparable.

Two things this argument does *not* use are worth stating, because the natural shorter proof uses both and is wrong. It never asserts $V^\star(W_1, T_a) \geq \mathrm{PASS}_M(W_1, T_a)$: the left side is a Bayes value under μ_{T_a}, the right a pass rate at the single realized convention, and optimality relates $V^\star$ only to the μ_{T_a}-averaged $\overline{\mathrm{PASS}}_M$ (Remark 1). And it never treats the measured crossover as the hypothesis: the crossover is the observable signature of $g_a, g_b < 1$, not a substitute for it. The caps remain essential in the converse direction: without them a PASS_M crossover is compatible even with Blackwell *equivalence*, since a fixed rule keyed to one signal's surface form can under-perform on the other, exactly the non-coherence phenomenon Theorem 3 verifies for (W_1, W_{1+}).

(b) Assumption 1 directly yields $\mathrm{PASS}_M(W_1, T_a) > \mathrm{PASS}_M(W_2, T_a)$ and the reverse at T_b, which is Definition 5's incomparability; the failure of the unconditional converse is witnessed by any fixed rule that violates monotonicity the optimal value obeys. □

Proof of Theorem 2

Proof. By incomparability, neither source garbles the other. By the contrapositive of Theorem 1, "not $W_1 \succeq_B W_2$" yields a task T_b (a prior plus bounded utility) with $V^\star(W_2, T_b) > V^\star(W_1, T_b)$, and "not $W_2 \succeq_B W_1$" yields a task T_a with $V^\star(W_1, T_a) > V^\star(W_2, T_a)$; both inequalities are *strict*, so ties do not arise. A query-independent φ assigns the single ordered pair $(\varphi(W_1), \varphi(W_2))$, fixing one total order of $\{W_1, W_2\}$ *independent of the task.* That fixed order agrees with the optimal-value order on at most one of T_a, T_b (they point opposite ways), hence is violated on the other. The PASS_M conclusion follows by replacing $V^\star$ with PASS_M via Assumption 1, which guarantees the realized rule reproduces both strict inequalities with margin γ. □

Proof of Proposition 2

Proof. For (a), $\varphi = g \circ \psi$ and $\psi(W_1, T_c) = \psi(W_2, T_c)$ force equal scores, while faithfulness on T_c would require a strict gap. For (b), $\mathrm{ov}(W_2, T_c) > \mathrm{ov}(W_1, T_c)$ with g strictly increasing gives $\varphi(W_2, T_c) > \varphi(W_1, T_c)$, while utility favours W_1 because T_c's PASS requires W_1's coordinate; the two orders disagree. Part (b) makes explicit the two things the informal reading of "saturated lexicon" presumes, strict monotonicity and overlap being the only feature read, without which no strict inequality follows. □

Proof of Proposition 1

Proof. Let G be the garbling with $\kappa_{W_2} = G \circ \kappa_{W_1}$. Any predictor $g \in \mathcal{V}$ acting on W_2's signal is realized on W_1's signal by the composite $g \circ G$, which lies in $\mathcal{V}$ by the closure hypothesis. Therefore the best $\mathcal{V}$-predictor under W_1 attains at least the predictive accuracy of the best under W_2, so its expected predictive risk is no larger; the risk-reduction relative to the common null signal is thus at least as large under W_1. The scalarity is immediate: $I_{\mathcal{V}}$ returns a single real number per source, so it induces a total order and, by Theorem 2, cannot rank a Blackwell-incomparable pair correctly for all tasks. □

Proof of Proposition 3

Proof. Let E be the event that all $2|\mathcal{D}|$ Clopper–Pearson intervals contain their true proportions. Each interval fails with probability at most $\alpha = \eta/(2|\mathcal{D}|)$ (exact coverage of Clopper–Pearson [Clopper and Pearson, 1934]), so by the union bound $\Pr[E^c] \leq 2|\mathcal{D}| \cdot \alpha = \eta$, giving $\Pr[E] \geq 1 - \eta$. On E, for every T we have $p_{W_2,T} \geq L_{W_2,T}$ and $p_{W_1,T} \leq U_{W_1,T}$, so $p_{W_2,T} - p_{W_1,T} \geq L_{W_2,T} - U_{W_1,T}$; taking $\max_T$ on both sides, $\sup_T[p_{W_2,T} - p_{W_1,T}] \geq L_{\mathcal{D}}(W_1, W_2)$. Thus on E (probability $\geq 1 - \eta$) the true sup is at least $L_{\mathcal{D}}(W_1, W_2)$, which is the claimed lower confidence bound; if it is positive, the true sup is positive, verifying $\widehat{\delta}_{\mathcal{D}}(W_1, W_2) > 0$. The reverse direction uses $L_{W_1,T}$ and $U_{W_2,T}$, which are endpoints of the *same* $2|\mathcal{D}|$ intervals already included in E; therefore both one-sided verifications are valid on the single event E, and their conjunction holds at $\geq 1 - \eta$ without a further Bonferroni factor. □

Proof of Proposition 4

Proof. Only two quantities enter $\overline{\Delta}_{T^\star}$: an upper endpoint on $p_{W_{1+},T^\star}$ and a lower endpoint on $p_{W_1,T^\star}$. The two arms are independent Bernoulli samples (separate completions). Let E_+ be the event $p_{W_{1+},T^\star} \leq U_{W_{1+},T^\star}$ and E_- the event $p_{W_1,T^\star} \geq L_{W_1,T^\star}$. By the exact one-sided coverage of Clopper–Pearson at noncoverage $\alpha = \eta/2$ per endpoint, $\Pr[E_+^c] \leq \eta/2$ and $\Pr[E_-^c] \leq \eta/2$, so by the union bound $\Pr[E_+ \cap E_-] \geq 1 - \eta$. On that event

$$\Delta_{T^\star} \;=\; p_{W_{1+},T^\star} - p_{W_1,T^\star} \;\leq\; U_{W_{1+},T^\star} - L_{W_1,T^\star} \;=\; \overline{\Delta}_{T^\star},$$

which is the claim; if $\overline{\Delta}_{T^\star} \leq -\tau$ then $\Delta_{T^\star} \leq -\tau$ at confidence $\geq 1 - \eta$. Note that the budget is spent on exactly the two endpoints used: a two-sided interval at level $\eta/2$ would place $\eta/4$ in each tail, leave two endpoints unused, and deliver $1 - \eta/2$ while widening the bound, which is why the allocation is stated per endpoint rather than per interval. For a post-hoc selected cell, replacing α by $\eta/(2|\mathcal{D}|)$ and union-bounding over the $2|\mathcal{D}|$ endpoints preserves $\geq 1 - \eta$ simultaneous coverage. □

Proof of Theorem 3

Proof. $W_{1+} \succeq_B W_1$ by deletion of the extra coordinate (a deterministic garbling, Lemma 1(a)); write G for that deletion, so $G \circ \kappa_{W_{1+}}(\cdot \mid s) = \kappa_{W_1}(\cdot \mid s)$ for every state s. The rule $\delta_M \circ G$ is then Blackwell-coherent on $T^\star$ in the sense of the theorem, and its realized pass rate on W_{1+} equals $\mathrm{PASS}_M(W_1, T^\star)$ at the realized state, since the two rules see the same signal law. The hypothesis $\mathrm{PASS}_M(W_{1+}, T^\star) < \mathrm{PASS}_M(W_1, T^\star)$ therefore says that δ_M on W_{1+} is strictly beaten by its own pruned version on the same input, so δ_M is not Blackwell-coherent on $T^\star$. Note what this argument does *not* use: it is the simulation construction behind (i)⇒(ii) of Theorem 1 applied state-conditionally, not an instance of statement (ii), which is Bayes-averaged and so says nothing about a single state (Remark 1). A monotone scalar ϕ assigns $\phi(W_{1+}) \geq \phi(W_1)$ (monotonicity), hence orders W_{1+} above W_1 and cannot reproduce the realized reversal. □

Relation to LLM-evaluation statistics

Remark 9 (Relation to LLM-evaluation statistics). The verification sits in a small but growing statistical-methodology literature for black-box model evaluation: paired-difference designs and clustered standard errors for evals [Miller, 2024], two-sample tests verifying behavioral differences between API-served models [Gao et al., 2025], and the recommendation to prefer exact frequentist intervals over CLT approximations when per-cell n is below a few hundred [Bowyer et al., 2025], which is precisely our regime ($n \in [20, 80]$ per cell) and precisely what the exact Clopper–Pearson construction provides.

Power and the role of n

Both verifications are driven by how close the arms sit to an endpoint. For a stark cell with empirical proportion at an endpoint ($\hat{p} \in \{0, 1\}$) the Clopper–Pearson interval becomes one-sided and clears 0 even at small n. It does not become a point: at $n = 40$ and two-sided level $\eta/8$, 0/40 gives [0, 0.119] and 40/40 gives [0.881, 1], and it is that 0.119 the crossing has to beat. The incomparability verification (6) is carried by exactly such cells (W_1 at $\approx$ 100%, W_2 at 0% and vice versa). The harm contrast (8) needs the gap between two arms to exceed τ after both intervals are paid for, so a mild collapse from a non-endpoint arm, as in the import-neutralized Haiku control (100% → 52%), is verified in sign but not at $\tau = 0.30$ at $n = 40$; that cell would clear at $n \approx 80$.

B Extended experimental record

Generality and controls: full protocols

Four follow-up experiments address three threat axes: source-pair generality, the context-length confound, and whether the anti-monotonicity collapse is curable by cheap prompting.

A second incomparable pair, different domain. We repeated the protocol with a second hand-built pair on a private `cache` module: V_1, the `cache.put(key, value, *, ttl)` call contract (argument order, required keyword-only `ttl`, a True/False "newly inserted" flag, a `CacheError` type), versus V_2, a key-normalization invariant (a fixed `kx7-` namespace prefix and an underscore rule). On Haiku at $n = 40$ the pair is again verified incomparable ($L_{\mathcal{D}} = +0.774/+0.496$, both directions clear of zero) with a clean double crossover (V_1 solves the contract and trap cells, V_2 solves key-normalization, all none-baselines floor at 0%), and the dominance control again verifies one-directionally. Anti-monotonicity reproduces in direction at $n = 40$ (the superset collapses the contract cell 80% → 28%, $\Delta_T = -0.52$); re-running the arms at $n = 80$ shows the collapse is stable (80% → 29%, $\Delta_T = -0.51$) with Clopper–Pearson upper bound −0.33, which clears the margin $\tau = 0.30$ fixed for the whole programme. The binding constraint is the V_1 ceiling itself (80%, versus pair-1's ∼100%), so the margin is met with little to spare; at $n = 40$ the same collapse gives −0.25 and does not clear, which is why the cell was re-run at $n = 80$. The $n = 80$ re-run covers the `cache_put_ok` cell alone, so its dominance control ($U_{\mathcal{D}} = 0.000$ with a verified strict win) is a statement about that cell rather than about the three-cell battery; the battery-wide control remains

Table A1: Per-cell PASS rates (%) for all four verified cells, by arm, all single-shot at $n = 40$. Every model was measured on every cell and arm. The encoding rows (`enc_amount`) carry the $W_2 > W_1$ direction of incomparability: on every model W_2 solves encoding while W_1 fails it, mirroring the API cells where W_1 solves and W_2 fails.

Model	Cell	`none`	W_1	W_2	W_{1+}
Haiku-4.5	`api_post_ok`	0	52	0	15
Haiku-4.5	`api_argorder`	2	100	0	5
Haiku-4.5	`enc_amount`	0	0	100	100
Haiku-4.5	`trap_store_wire`	0	90	0	92
Sonnet-4.6	`api_post_ok`	0	100	0	2
Sonnet-4.6	`api_argorder`	0	100	0	0
Sonnet-4.6	`enc_amount`	0	0	98	80
Sonnet-4.6	`trap_store_wire`	0	100	0	65
Opus-4.8	`api_post_ok`	0	100	0	12
Opus-4.8	`api_argorder`	40	100	0	5
Opus-4.8	`enc_amount`	0	0	98	100
Opus-4.8	`trap_store_wire`	0	100	0	100
GPT-5.5	`api_post_ok`	5	100	0	0
GPT-5.5	`api_argorder`	75	100	0	0
GPT-5.5	`enc_amount`	55	0	100	100
GPT-5.5	`trap_store_wire`	0	100	0	100
DeepSeek-V4-Pro	`api_post_ok`	0	75	0	0
DeepSeek-V4-Pro	`api_argorder`	2	68	0	5
DeepSeek-V4-Pro	`enc_amount`	0	0	68	65
DeepSeek-V4-Pro	`trap_store_wire`	0	98	0	98
Kimi-K2.6	`api_post_ok`	0	100	0	0
Kimi-K2.6	`api_argorder`	45	100	0	0
Kimi-K2.6	`enc_amount`	0	0	95	92
Kimi-K2.6	`trap_store_wire`	0	100	0	100

the $n = 40$ one. The relevance trap for this pair was a null, its prompt vocabulary leaned toward V_1 ($\varphi(V_1) = 0.39 > \varphi(V_2) = 0.25$), so relevance *agreed* with PASS, so the strong mis-rank remains a pair-1 result. The contribution of the second pair is therefore *generality of the incomparability and dominance results to a new domain*, not a second mis-rank.

A character-matched padding control. Anti-monotonicity compares W_1 against the strictly longer W_{1+}, and added input length alone is known to degrade performance [Du et al., 2025], so the collapse could in principle be a token-count effect rather than a signal effect. To isolate this we added a padding arm W_{pad}: filler text in the same "team context (private)" register as W_2, placed in the same position W_2 occupies in W_{1+} (padding first, W_1 last), character-matched to W_{1+} within 13 characters, but carrying no coordinate any verifier checks (office-logistics content). On single-shot Haiku at $n = 40$ on the two collapse cells, padding is free while the equal-length dominated signal is catastrophic: on `api_argorder`, W_1 98%, W_{pad} 100%, W_{1+} 10%; on `api_post_ok`, W_1 50%, W_{pad} 65%, W_{1+} 10%; and the same-run interference verification clears on `api_argorder` ($\Delta_T = -0.88$, upper bound -0.67). The anti-monotonicity harm is therefore attributable to W_2's signal content interacting with the fixed rule, not to added tokens. One residual axis is not isolated by this control:

the padding is off-topic for the `ledger` module while W_2 is on-topic, so "topical but verifier-irrelevant" text remains a smaller, untested middle case.

A routing-instruction ablation: the cure presupposes the diagnosis. If the collapse reflects convention confusion, an explicit routing note might prevent it, and if a one-line prompt fully cured the harm, its practical significance would shrink. We measured this directly: the arm W_{1+}^{instr} appends to W_{1+} a single routing sentence stating which convention serves which task family ("for tasks that call the ledger API use only the API call contract; the wire-encoding invariant applies only to encode/decode tasks"). On single-shot Haiku at $n = 40$ the instruction recovers most, but not all, of the collapse: on `api_argorder`, $2\% \to 80\%$ (W_1 alone: 100%), and on `api_post_ok`, $22\% \to 68\%$ (W_1 alone: 50%); both recoveries over W_{1+} are large (non-overlapping exact 95% intervals), while on the carrying cell the instructed superset still sits below W_1 alone (80% vs. 100%). That residual gap is verified negative by the same two-arm contrast used throughout, $\Delta_T = -0.20$ with upper bound -0.03, but it falls far short of the harm margin $\tau = 0.30$: the instruction removes the collapse without closing the gap. We pre-commit to the two readings jointly. First, the harm is substantially mediated by convention interference that explicit routing knowledge prevents, so pipelines that can attach per-task routing labels have a cheap mitigation. Second, and decisively for the paper's thesis: *writing the routing sentence requires already knowing which source is decisive for which task family*, which is precisely the coordinate structure the partial order formalizes and the deficiency verification measures; the cure is an injection of selection-time coordinate knowledge into the prompt, not a scalar score, and no query-independent scalar can supply it (Theorem 2). Formally, δ_M with the routing note is a *different, more Blackwell-coherent rule*; the verified falsification of Theorem 3 concerns the base rule and is unaffected, and even under the instructed rule, keeping the dominated signal is at best equal and directionally worse than dropping it, so the selection-level conclusion, keep the non-dominated set rather than forcing the superset, survives the ablation intact.

A structured-context ablation: labeled XML segmentation mitigates only weakly. To separate "the model cannot segment the two conventions" from "the model cannot route them to task families," the arm W_{1+}^{xml} wraps the same two conventions, in the same order as W_{1+}, in labeled XML sections (`<wire_encoding_invariant>`, `<api_call_contract>`) with no routing statement. On single-shot Haiku at $n = 40$ per cell, structure alone mitigates weakly: `api_argorder` 45% (vs. W_{1+} 2%, W_{1+}^{instr} 80%, W_1 100%) and `api_post_ok` 25% (vs. 12%, 68%, 57%). The structure-only mitigation is separated from the raw superset on the carrying cell (exact 95% intervals $[0.27, 0.59]$ vs. $[0.001, 0.13]$) but remains far below W_1 alone; on `api_post_ok` it is not separated from the raw superset. Protocol note: the W_{1+}^{xml} arm was measured in a clean arm-only re-run (its arm in the first run was invalidated by a transport-failure window and scored no completions; the re-run had zero transport errors), so its comparison against same-day baseline arms is across runs rather than within one run. The resulting mitigation hierarchy, raw superset < structure < routing < dropping the dominated signal, is consistent with the operative variable being convention-to-task routing rather than segmentation alone; as in §8.3 this localization is an interpretation of the behavioural contrasts, not a verified claim, and it rests on one model at $n = 40$ with the structure arm scored in a separate run.

C Reproducibility and protocol

All reported numbers come from real model calls whose outputs are checked by executable verifiers; there are no modeled or aspirational figures. The complete artifact bundle (the measurement harness, the executable verifiers, and every per-model run log and result JSON cited below) is available at https://github.com/abilous-ti/blackwell-llm-context. The harness (`harness/measure_blackwell.py`) is Python-standard-library only (no third-party dependencies; the Clopper–Pearson endpoints use a stdlib regularized incomplete beta), is cross-OS, and implements the surrogate $\widehat{\delta}_{\mathcal{D}}$, the uniform incomparability verification (`certify_incomparable`, `clopper_pearson`), the dominance control (`certify_dominance`), the interference verification, the sample-size helpers (`required_n`, `required_n_stark`), and the lexical-relevance scalar (`lexical_relevance`). It includes launchers for the Anthropic CLI (`claude -p`), the Azure Responses API, and OpenAI-compatible chat completions; the launcher auto-detects API style from the endpoint, and all API keys are read from the environment (never written to a file).

Protocol. The verifier stubs a fake `ledger` module that accepts *only* the exact private contract (it rejects wrong positional arg order, a missing `memo=` keyword, and the wrong exception type); return code 0 is PASS. The four verified cells are `api_post_ok`, `api_argorder` (`memo` arrives via a parameter, forcing the keyword-only fact), `enc_amount`, and `trap_store_wire`; the arms are $\{\text{none}, W_1, W_2, W_{1+}\}$, with W_{pad} (character-matched neutral filler $+\!\!+\, W_1$) added for the padding control. All six models are queried in a single-shot completion (one call, no tools, code extracted from the returned text), and PASS is measured over $n = 40$ i.i.d. completions per cell at $\eta = 0.10$.

Decoding, seeding, and retries. All runs use the provider's default decoding configuration: the harness sets no explicit temperature, top-p, or seed (the Claude CLI and the Azure/OpenAI-compatible launchers are invoked without sampling overrides), so the n draws per cell are independent samples from each provider's default decoder, each a fresh single-turn call with a 200-second timeout. This describes every run behind the six-model tables. Two reported runs sit outside it, both on the agentic Claude CLI path (up to six turns per attempt) rather than the single-shot one, and both identifiable from their run logs, which lack the `[SINGLE-SHOT]` marker every single-shot run carries. The first is the order control of §7.4.3, an earlier $n = 80$ Haiku run, reported as an auxiliary control and entering no table of the six-model record. The second is the entire second source pair (§7.5), measured agentically at $n = 40$ and $n = 80$. Its role is generality, so the regime difference is worth stating plainly: what that pair shows is that incomparability and the superset collapse reappear in another domain *under the agentic harness*, not that they reappear under the single-shot one. The single-shot evidence for both phenomena is the six-model record on the first pair.

Four arm-only re-measurements were originally produced by ad-hoc scripts that were not kept, so their regime could be asserted but not recovered from a run log. All four have since been re-measured with `harness/diag/rerun_arm.py`, which calls the same `run_one` the six-model tables are built from and writes a log recording the regime. Three reproduced their published values closely: the Opus W_2/`trap_store_wire` and Haiku W_1/`enc_amount` cells returned 0/40 exactly as before, and the structured-context arm moved by one draw on one cell. The Opus W_{1+} arm moved further, and deeper: 38% → 12% on `api_post_ok` and 15% → 5% on `api_argorder`. This manuscript reports the re-measurements throughout, so every cited run now has a log naming its regime; `harness/diag/check_regimes.py` lists them with the evidence for each. The between-run movement on the Opus arm is itself a data point for §8.4: repeated measurement of the same cell months apart shifts by more than the nominal interval width. Transport-level failures (empty or zero-token results) are retried up to three times with quadratic backoff; a draw that still fails is scored as a failure, never as a pass. Where a transport-outage window hit a whole arm or a single cell, we re-measured it cleanly in isolation with the retry harness and report only the clean measurement: the Opus W_{1+} arm; the Opus W_2/`trap_store_wire` cell, roughly 18 of whose 40 first-run draws were outage-scored (inferred from per-draw cost); and the Haiku W_1/`enc_amount` cell, whose first run predated the retry wrapper. One cell resisted re-measurement for longer than the others: GPT-5.5 W_2/`trap_store_wire` returned HTTP 500 after retries during the original run and no GPT-5.5 deployment was available on the resource at first revision, so for one revision cycle its reported 0/40 was carried with outage-scored draws in it. A GPT-5.5 deployment was subsequently provisioned and all four GPT-5.5 cells were re-run single-shot at $n = 40$ per arm with retention and zero transport failures. The re-run reproduces the published grid exactly in all eight W_1/W_2 entries, the disputed cell included (W_2/`trap_store_wire` 0/40 clean), so every cell in the coverage family is now a valid binomial sample and the reported n is the number of real draws throughout. The GPT-5.5 rows of Table 6 come from that run.

Artifacts. Per-model run logs and result JSONs are in the artifact repository (https://github.com/abilous-ti/blackwell-llm-context) under the pattern `blackwell_*.{log,json}`: the six single-shot $n = 40$ model runs (`blackwell_haiku_ss_n40`, `blackwell_sonnet_ss_n40`, `blackwell_opus_ss_n40` with its arm-only re-run `blackwell_opus_ss_W1plus_v2` and the clean single-cell re-measurements `blackwell_opus_ss_W2_trap_v2` and `blackwell_haiku_ss_W1_enc_v2`, all produced by `harness/diag/rerun_arm.py`, `blackwell_gpt55_n40`, `blackwell_deepseek_n40`, `blackwell_kimi_n40`), and the follow-up controls `blackwell_pad_n40b` (character-matched padding, 400 calls), `blackwell_order_ss_n40` (order control, reversed concatenation, single-shot, 400 calls), `blackwell_pair2_n40` (second source pair; the reported $L_{\mathcal{D}}$, $U_{\mathcal{D}}$ and Δ_T all come from this file) and `blackwell_pair2_n80` (second source pair), `blackwell_instr_n40` (routing-instruction ablation), `blackwell_xml_n40` / `blackwell_xml_arm_v2` (structured-context ablation), `reranker_trap_n30` (LLM listwise reranker probe; `harness/measure_reranker_trap.py`), and `dense_trap.json` (dense/cross-encoder probe, deterministic local scoring of open-weight checkpoints at zero API cost; `harness/measure_dense_trap.py`). The full per-cell record is in `docs/EXPERIMENT-BLACKWELL.md` and the theory hardening (formalization and proof sketches) in `docs/BLACKWELL.md`.

Compute and cost. Total measured Anthropic spend was approximately \$130: the single-shot Haiku, Sonnet, and Opus runs (Opus the costliest, plus its arm-only re-run) and the follow-up controls (padding, second pair, routing, structured-context). The non-Anthropic runs (GPT-5.5, DeepSeek, Kimi) used Azure quota; the Azure API returns no cost field, so this figure is the Anthropic spend, not the total across all six models. The dense/cross-encoder probe ran locally on open weights at zero API cost.

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